Camassa–Holm peakons under momentum transport noise. Part I: Global well-posedness and probabilistic stability

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ABSTRACT

We study the stochastic Camassa–Holm model arising from Holm's variational principle (2015), as developed by Holm and Tyranowski (2016) [27, 30]. For momentum transport noise generated by finitely many spatially dependent C_b^3 fields, global probabilistically strong solutions are constructed and pathwise uniqueness is established for arbitrary nonnegative finite initial momentum, allowing atoms without spatial moment assumptions. Within this class, quantitative stability estimates for the positive single-peakon family with evolving height and position depend only on the initial H^1 velocity error and allow arbitrarily large perturbation momenta. These estimates follow from a nonnegative defect of the quadratic and cubic CH integrals. Representing this defect by a positive Green operator gives bounds for its first two noise derivatives relative to the defect itself, including near rank one. The resulting multiplicative stochastic balance yields exit-probability bounds and mean-exit lower bounds, so for $\varepsilon > 0$ an initial error δ gives probabilistic stability on intervals proportional to $\varepsilon^{-2}\log(1/\delta)$ with an explicit admissible proportionality constant. Exact solutions and reproducible particle computations illustrate how the peakon shape persists while the amplitude evolves.

Part II [41] establishes probabilistic stability of positive multipeakon sets and H^1 continuous dependence at a specified multipeakon driven by the same Brownian motions.

1. Introduction

1.1. Formulation of the problem

The Camassa–Holm (CH) equation was introduced as a model for shallow water waves and admits a bi-Hamiltonian structure and an isospectral formulation [8, 9, 21]. Its dispersionless form on the real line is

$$m_t + um_x + 2u_x m = 0, \quad m = u - u_{xx}, \quad u = G * m, \quad G(x) = \tfrac{1}{2}e^{-|x|}.$$

The peaked solitary waves $u(t,x) = ce^{-|x-ct-q_0|}$, $c > 0$, belong to $H^1(\mathbb{R})$ and have momentum $2c\delta_{ct+q_0}$. These waves are therefore naturally described by nonnegative momentum measures. More generally, global weak solutions exist for nonnegative finite momentum, covering both peakons and velocities with diffuse momentum [16]. In this class, the quadratic and cubic conserved quantities

$$E(u) = \int_{\mathbb{R}} (u^2 + u_x^2)\,\mathrm{d}x, \qquad \mathcal{F}(u) = \int_{\mathbb{R}} (u^3 + uu_x^2)\,\mathrm{d}x$$

combine with a sharp peakon inequality to establish deterministic orbital stability for a peakon of prescribed height [17]. These properties lead us to study the persistence of peakon shape when momentum is transported by a stochastic flow.

Holm's stochastic variational principle [27] introduces this transport through the Stratonovich differential

$$\mathrm{d}\chi_t(x) = u_t(x)\,\mathrm{d}t + \varepsilon \sum_{k=1}^{N} \xi_k(x) \circ \mathrm{d}W_t^k,$$

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where $W^1, \dots, W^N$ are independent Brownian motions and the deterministic fields ξ_k describe the spatial structure of the noise. For the CH model, the H^1 Lagrangian is

$$\ell(u) = \frac{1}{2}\int_{\mathbb{R}} (u^2 + u_x^2)\,\mathrm{d}x, \qquad m = \frac{\delta \ell}{\delta u} = u - u_{xx}.$$

For smooth velocities, the Euler–Poincaré equation transports the momentum as the one-form density $m\ \mathrm{d}x \otimes \mathrm{d}x$ [28]. Since

$$\mathcal{L}_v(m\ \mathrm{d}x \otimes \mathrm{d}x) = (vm_x + 2v_x m)\ \mathrm{d}x \otimes \mathrm{d}x,$$

the stochastic transport law $\mathrm{d}(m\ \mathrm{d}x \otimes \mathrm{d}x) + \mathcal{L}_{\mathrm{d}\chi_t}(m\ \mathrm{d}x \otimes \mathrm{d}x) = 0$ expands into the stochastic CH equation [27, 30], written with the sign convention of [18] as

$$\mathrm{d}m + (um_x + 2u_x m)\,\mathrm{d}t + \varepsilon \sum_{k=1}^{N} (\xi_k m_x + 2\xi_k' m) \circ \mathrm{d}W_t^k = 0, \quad m = u - u_{xx}. \tag{1.1}$$

The parameter $\varepsilon \geq 0$ separates the noise amplitude from its spatial structure: for each fixed ε, replacing ξ_k by $\widetilde{\xi}_k = \varepsilon \xi_k$ incorporates the amplitude into the fields, as in Holm's notation. Keeping ε explicit records the dependence of the stability estimates on noise strength and includes deterministic CH at $\varepsilon = 0$. Throughout, the noise fields satisfy $\xi_k \in C_b^3(\mathbb{R})$ and the initial momenta are deterministic, nonnegative and finite, with the weak formulation specified in Definition 2.1. The coefficient 2 combines the one-form and density contributions to transport; in velocity variables, with $\mathcal{A} = 1 - \partial_x^2$, it yields

$$\mathcal{A}^{-1}(\xi m_x + 2\xi' m) = \xi u_x + \mathcal{A}^{-1}(2\xi' u + \xi'' u_x).$$

This transport structure preserves the single-peakon form while allowing its height and position to evolve. More precisely, the height and position of $u_t = a_t e^{-|\cdot - q_t|}$ solve

$$\mathrm{d}q_t = a_t\,\mathrm{d}t + \varepsilon \sum_k \xi_k(q_t) \circ \mathrm{d}W_t^k, \qquad \mathrm{d}a_t = -\varepsilon a_t \sum_k \xi_k'(q_t) \circ \mathrm{d}W_t^k.$$

These dynamics motivate comparison with the family

$$\mathcal{P} = \{ae^{-|\cdot - q|} : a > 0,\ q \in \mathbb{R}\}, \qquad d_{\mathcal{P}}(u) = \inf_{\varphi \in \mathcal{P}} \|u - \varphi\|_{H^1}.$$

Our stability problem concerns solutions whose initial velocities are close to a peakon in H^1 and whose momenta may contain both atomic and diffuse parts. Comparison with $\mathcal{P}$ allows the peakon height and position to evolve, as illustrated by the exact solutions in Section 6.

To study these perturbations, we establish global probabilistically strong solutions and pathwise uniqueness for every nonnegative finite initial momentum. The construction admits atoms and imposes no spatial moment condition, so exact peakons and their general positive-momentum perturbations are treated within the same global solution theory. For these global solutions, we prove quantitative probabilistic stability of $\mathcal{P}$ in terms of the initial H^1 velocity error, allowing perturbations with arbitrarily large total momentum. The argument uses a nonnegative defect of the quadratic and cubic CH integrals, whose stochastic drift and noise coefficients are controlled relative to the defect itself. The resulting balance gives explicit exit-probability bounds and lower bounds for mean exit times. In particular, for $\varepsilon > 0$, an initial error δ from a peakon of prescribed height yields persistence on intervals $a\varepsilon^{-2}\log(1/\delta)$: for an explicit noise-dependent range of $a > 0$, the probability of leaving any fixed H^1 neighborhood of $\mathcal{P}$ by this time tends to zero as $\delta \downarrow 0$. The precise assumptions and conclusions are stated in Section 1.3, while Part II [41] develops the corresponding theory for positive multipeakon sets and comparison with a specified multipeakon driven by the same Brownian motions.

1.2. Related works

For deterministic CH, a sign condition on sufficiently regular momentum yields global existence [14], whereas general solutions may undergo wave breaking with bounded amplitude [15]. Bressan and Constantin [5, 6] constructed conservative and dissipative continuations, and Holden and Raynaud [26] developed a Lagrangian construction of

conservative solutions. These theories account for concentration of energy at wave breaking. For nonnegative finite momentum, Constantin and Molinet [16] constructed global weak solutions; positive particle approximations also lead to this class [25, 36].

The deterministic stability mechanism is particularly relevant here. Constantin and Strauss [17] combined conservation of the quadratic and cubic functionals with a sharp inequality at the wave maximum to prove single-peakon orbital stability in H^1. Lenells [32] proved periodic peakon stability. Under spatially varying momentum noise, the quadratic and cubic functionals evolve stochastically, and the relative estimates established below control their defect from the peakon relation. Other deterministic continuous-dependence theories use distances adapted to the conservative flow: optimal transportation in Bressan and Fonte [7], a Lipschitz metric on the line in Grunert et al. [23], and a Wasserstein construction in Carrillo et al. [10]. These metrics describe the conservative state, including an energy measure when necessary. The present exit estimate is formulated directly in terms of the initial H^1 velocity error.

For stochastic momentum transport, Holm and Tyranowski [30] derived singular-solution equations, and Crisan and Holm [18] proved preservation of the peakon ansatz and wave breaking under their stated hypotheses. These results describe exact singular solutions; perturbation stability also requires dynamics admitting non-atomic momentum. Albeverio et al. [1, Theorem 2.2] proved local strong well-posedness for momentum in $H^{2,q}(\mathbb{R})$, $1 < q < \infty$, with one Brownian driver. Their first-order convection operator includes full momentum transport; the Doss–Sussman transformation and Kato argument assume a C_b^4 transport field. The global theory below admits arbitrary nonnegative finite measures, including atoms.

Other stochastic CH theories treat different noise operators or solution classes. Chen et al. [13] studied additive noise. Tang [40] proved local pathwise well-posedness for periodic Sobolev data under multiplicative noise and, in the linear case, studied global existence, blow-up probabilities and a pathwise dissipative effect on periodic peakons. Ren et al. [38] treated distribution- and path-dependent noise in a general nonlinear SPDE framework. Galimberti et al. [22] established global periodic dissipative martingale solutions with velocity transport noise; the identity in Section 1.1 shows how this operator differs from full momentum transport for spatially varying fields. Chen et al. [11] constructed global $H^1(\mathbb{R})$ martingale solutions under linear multiplicative noise, and Luo et al. [33] obtained periodic H^1 weak martingale solutions under Itô linear multiplicative noise. Chen et al. [12, Theorem 1.1] proved modulation and exit estimates starting from an exact smooth solitary wave, under Marcus jump velocity transport noise in a linearly dispersive CH model. Bendall et al. [2] investigated weak stochastic peakons and their formation numerically, while Holm et al. [29] compared stochastic Hamiltonian CH models with different conservation properties. The exit estimates below quantify persistence near the peakon family over explicit time intervals. Deterministic asymptotic stability [34] addresses the long-time approach to a peakon; the extension to certain sign-changing perturbations [35] requires conditions on the relative locations of positive and negative momentum.

This paper combines a global pathwise solution theory for nonnegative finite momentum measures with a relative quadratic–cubic defect estimate to derive probabilistic stability of the single-peakon family from initial H^1 proximity. Atoms are admitted, and neither a spatial moment assumption nor a uniform bound on the perturbations' total momenta is needed. Part II [41] uses this solution theory to study positive multipeakon sets through exterior powers and positive atomic reconstruction, and to compare a perturbed solution with a specified multipeakon driven by the same Brownian motions. Its set estimate includes the single-peakon case; the global solution theory and the quadratic–cubic mechanism are established here independently.

1.3. Main results

The stability estimates are formulated within a global well-posedness theory for (1.1) in $\mathcal{M}_+(\mathbb{R})$, the space of nonnegative finite Radon measures on $\mathbb{R}$. The space $\mathcal{M}_+(\mathbb{R})$ carries the narrow topology, defined by convergence against bounded continuous functions [4, Section 8.1], which in particular entails convergence of the total momentum $M(m) = m(\mathbb{R})$. Fix $\varepsilon \geq 0$, independent Brownian motions and fields $\xi_k \in C_b^3(\mathbb{R})$, $1 \leq k \leq N$. The weak formulation and moment bounds are specified in Definition 2.1, and the choice $G'(0) = 0$, the average of the one-sided limits, determines the kernel derivative at atoms. Within this class, a strong solution is understood in the probabilistic sense: it is constructed on the prescribed probability space with the given Brownian motions.

Theorem 1.1 (Global existence and pathwise uniqueness)**.** *For every deterministic $m_0 \in \mathcal{M}_+(\mathbb{R})$, there is a global solution in the sense of Definition 2.1, adapted to the completed Brownian filtration on any prescribed Brownian*

probability space. Any two such solutions on the same space with the same m_0 and W are indistinguishable. The solution is also unique in law. It has the representation

$$m_t = (q_t)_\#(p_t\, m_0), \quad q_0(x) = x, \quad p_0(x) = 1. \tag{1.2}$$

The map q_t is strictly increasing on all of $\mathbb{R}$, and for every $T < \infty$, almost surely,

$$\sup_{t\le T}\|q_t - \mathrm{id}\|_\infty < \infty, \quad \Big\| \sup_{t\le T}\left(p_t + p_t^{-1}\right)\Big\|_{L^\infty(m_0)} < \infty.$$

The position equation holds almost surely for each fixed initial label, and both characteristic equations hold simultaneously for m_0-almost every label:

$$\begin{aligned} \mathrm{d}q(x) &= u(q(x))\,\mathrm{d}t + \varepsilon \sum_k \xi_k(q(x))\circ \mathrm{d}W^k, \\ \mathrm{d}p(x) &= -p(x)\bar{u}_x(q(x))\,\mathrm{d}t - \varepsilon p(x) \sum_k \xi_k'(q(x))\circ \mathrm{d}W^k. \end{aligned} \tag{1.3}$$

Here $\bar{u}_x = G' * m$ is defined using $G'(0) = 0$.

Remark 1.2. The global solution in Theorem 1.1 is obtained by combining the weak construction in Section 3 with pathwise uniqueness and the strong-solution construction in Section 4 on the prescribed Brownian space. For solutions driven by the same Brownian motions, Corollary 4.2 further gives convergence in probability on every finite interval under narrow convergence of the initial momenta, both for the momentum paths in the narrow topology and for the velocities in $C([0,T];H^1)$. In particular, it identifies the limit of the full positive particle approximation on that space.

To study the stability of peakons, perturbations with arbitrary total momentum are measured by their initial H^1 velocity error. The required estimates follow from stochastic balance identities valid for every solution of Definition 2.1, independently of pathwise uniqueness. To express these identities, for $u = G * m$ define the quadratic and cubic functionals

$$E(u) = \|u\|_{H^1}^2 = \iint_{\mathbb{R}^2} G(x-y)m(\mathrm{d}x)m(\mathrm{d}y), \quad \mathcal{F}(u) = \int_{\mathbb{R}} (u^3 + uu_x^2)\,\mathrm{d}x. \tag{1.4}$$

With $E(m) = E(G * m)$ in momentum variables, define the defects

$$\mathcal{J}(u) = \frac{\sqrt{2}}{3}E(u)^{3/2} - \mathcal{F}(u), \quad \mathcal{Z}(u) = \sqrt{E(u)}\mathcal{J}(u).$$

These defects are defined continuously at $u = 0$, and Proposition 5.1 establishes their nonnegativity on positive-momentum velocities together with the associated bound for $d_{\mathcal{P}}(u)$. The stochastic estimates use the nonnegative constants C_Z, V_Z, defined explicitly in (1.9) in terms of the noise fields, together with $\lambda_Z = \varepsilon^2 C_Z$. We also write $[a]_+ = \max\{a, 0\}$.

Theorem 1.3 (Stability under H^1 perturbations and exit estimates)**.** *Let m_0 be a deterministic nonzero nonnegative finite measure, and let m be any solution of Definition 2.1. For $\eta > 0$, set*

$$\tau_\eta = \inf\{t \ge 0 : d_{\mathcal{P}}(u_t) \ge \eta\}, \quad b_\eta = \frac{\eta^4}{4\sqrt{2}}.$$

(i) Initial H^1 perturbations. *For any $c > 0, q_0 \in \mathbb{R}$ and $\delta \ge 0$, $\|u_0 - ce^{-|\cdot - q_0|}\|_{H^1} \le \delta$ implies that for every finite T*

$$\mathbb{P}(\tau_\eta \le T) \le \min\left\{1,\, 32(\sqrt{2}c + \delta)^2 e^{C_Z\varepsilon^2 T}\frac{\delta^2}{\eta^4}\right\}. \tag{1.5}$$

No bound or proximity condition on $M(m_0)$ is imposed.

(ii) Defect balance and exit probabilities. *The process $e^{-\lambda_Z t}\mathcal{Z}(u_t)$ is a nonnegative continuous supermartingale. For every finite T, this supermartingale yields*

$$\mathbb{P}(\tau_\eta \le T) \le \min\left\{1, \frac{e^{\lambda_Z T}\mathcal{Z}(u_0)}{b_\eta}\right\}. \tag{1.6}$$

When $\mathcal{Z}(u_0) > 0$ and $\varepsilon^2 V_Z T > 0$, one also has

$$\mathbb{P}(\tau_\eta \le T) \le \exp\left\{-\frac{[\log(b_\eta/\mathcal{Z}(u_0)) - \lambda_Z T]_+^2}{2\varepsilon^2 V_Z T}\right\}. \tag{1.7}$$

(iii) Mean exit times and invariance. *If $0 < \mathcal{Z}(u_0) < b_\eta$ and $\lambda_Z > 0$, then*

$$\mathbb{E}\tau_\eta \ge \lambda_Z^{-1} \log \frac{b_\eta}{\mathcal{Z}(u_0)}. \tag{1.8}$$

The expectation may be infinite. If $\mathcal{Z}(u_0) = 0$, the solution remains in $\mathcal{P}$ almost surely.

Remark 1.4. With $L_{j,k} = \|\xi_k^{(j)}\|_\infty$, the constants in Theorem 1.3 are given by

$$\begin{aligned} &\kappa_{1,k} = 2(L_{1,k} + L_{3,k}), \quad \kappa_{2,k} = 4(L_{1,k} + L_{3,k})(L_{0,k} + L_{1,k} + L_{2,k}), \\ &C_Z = \frac{1}{2}\sum_k (80\kappa_{1,k}^2 + 4\kappa_{2,k}), \quad V_Z = 16\sum_k \kappa_{1,k}^2, \quad \lambda_Z = \varepsilon^2 C_Z. \end{aligned} \tag{1.9}$$

Since C_Z and V_Z depend only on the spatial noise fields, the bound in Theorem 1.3(i) is independent of total momentum and requires no energy stopping time. For fixed $\theta > 0$ and $T = \theta\varepsilon^{-2}$, this bound tends to zero as $\delta \downarrow 0$, uniformly in $\varepsilon > 0$ and all admissible perturbation momenta.

Remark 1.5. Theorem 1.3(i) converts the initial H^1 error into an exit-probability bound for the global dynamics of Theorem 1.1, allowing atomic and diffuse perturbation momenta without a uniform bound on total momentum. It therefore extends the exact stochastic peakon descriptions in [30, 18] to perturbations of the evolving peakon family under finitely many C_b^3 Brownian transport fields. In this comparison, both height and position vary, while the deterministic orbital stability of Constantin and Strauss [17] concerns a peakon of fixed height.

Assertions (ii)–(iii) explain the mechanism behind (i): relative drift and noise bounds for $\mathcal{Z}$ yield exit estimates, lower bounds for mean exit times and preservation of $\mathcal{P}$ at zero defect. Because the balance identities hold for every solution of Definition 2.1, these estimates apply independently of pathwise uniqueness. The resulting bounds concern peaked profiles and general initial H^1 perturbations under full momentum transport, whereas the modulation and exit estimates of Chen et al. [12] start from an exact smooth solitary wave under Marcus jump velocity transport noise.

To quantify how the interval of probabilistic stability grows as the initial H^1 error decreases, we apply Theorem 1.3(i) on a logarithmic time scale and obtain the following corollary.

Corollary 1.6 (Logarithmically growing time horizons)**.** *Assume $C_Z > 0$ and fix $c, \eta > 0$ and $0 < a < 2/C_Z$. For $0 < \delta < 1$, let m_0^δ be a deterministic nonzero nonnegative finite momentum with*

$$\|G * m_0^\delta - ce^{-|\cdot - q_\delta|}\|_{H^1} \le \delta, \quad q_\delta \in \mathbb{R},$$

and let τ_η^δ denote the corresponding exit time from the η-neighborhood of $\mathcal{P}$. For every $\varepsilon > 0$, set $T_\delta = a\varepsilon^{-2}\log(1/\delta)$. Then

$$\mathbb{P}(\tau_\eta^\delta \le T_\delta) \le \min\left\{1, \frac{32(\sqrt{2}c+1)^2}{\eta^4}\delta^{2-aC_Z}\right\} \longrightarrow 0 \quad (\delta \downarrow 0). \tag{1.10}$$

The convergence is uniform in $\varepsilon > 0$ and q_δ and requires no uniform bound on $M(m_0^\delta)$.

Proof. Substituting $T = T_\delta$ in (1.5) gives $e^{C_Z\varepsilon^2 T_\delta} = \delta^{-aC_Z}$. Since $\delta < 1$ and $2 - aC_Z > 0$, the conclusion follows. □

Remark 1.7. Corollary 1.6 applies the estimate in Theorem 1.3(i) on an interval whose length grows as the initial H^1 error decreases. For fixed $c, \eta > 0$ and $C_Z > 0$, the restriction $0 < a < 2/C_Z$ balances the quadratic initial error against exponential growth in the defect estimate, ensuring $\delta^{2-aC_Z} \to 0$. For each fixed $\varepsilon > 0$, the corresponding horizon T_δ tends to infinity as $\delta \downarrow 0$, while the exit-probability bound remains uniform in $\varepsilon > 0$, the reference position and all admissible perturbations, including those with arbitrarily large finite total momentum.

1.4. Main ideas of the proof

The proof combines a measure-valued existence theory with a stability estimate that remains effective near the peakon family. Atomic momenta and the jump of G' require a weak formulation compatible with ordered particle dynamics, while a coercive weight yields compactness without spatial moment assumptions. In the stability argument, the quadratic and cubic CH integrals evolve stochastically, while small H^1 errors can coexist with large total momentum. The required estimates therefore bound the drift and noise coefficients of the nonnegative defect $\mathcal{Z}$ relative to the defect itself, including near zero defect and rank-one Green operators. The argument proceeds through the following five steps.

(1) **Constructing global weak solutions.** Moment estimates for the total momentum, together with positivity and order preservation, ensure that the particle approximations are global. A slowly growing coercive weight gives tightness without a spatial moment assumption. Identifying the limiting equation jointly with its Brownian driver then gives a solution satisfying Definition 2.1.

(2) **Obtaining pathwise uniqueness and strong solutions.** In the weak solution class of Definition 2.1, stochastic-flow theory [31] and the Kunita–Itô–Wentzell formula for differential forms [3] remove the transport noise. A double-commutator estimate justifies this transformation for measures under C_b^3 regularity. The resulting equation has ordered characteristics, which permit comparison across the jump of G' at zero and yield pathwise uniqueness. Combined with weak existence, this gives the global probabilistically strong dynamics of Theorem 1.1.

(3) **Measuring the peakon error by a coercive defect.** For positive-momentum velocities, the quadratic–cubic defect controls the distance to the full peakon family through

$$d_{\mathcal{P}}(u)^4 \le 4\sqrt{2}\mathcal{Z}(u).$$

At a peakon, the first variation of $\mathcal{J}$ vanishes, so a bound on its second derivative gives a quadratic estimate in the initial H^1 velocity error. Together, these bounds reduce Theorem 1.3(i) to controlling $\mathcal{Z}$, with constants independent of total momentum.

(4) **Deriving relative bounds from the Green operator.** The noise derivatives of the defect are estimated through the positive operator on $L^2(m)$ given by

$$(T_m f)(x) = \int_{\mathbb{R}} e^{-|x-y|/2} f(y) m(\mathrm{d}y), \qquad \Phi(T) = (\operatorname{tr} T^2)^{3/2} - \operatorname{tr} T^3.$$

The identity $6\mathcal{J}(u) = \Phi(T_m)$ relates the peakon defect to rank one. More precisely, for a nonzero positive trace-class operator with eigenvalues $\lambda_1 \ge \lambda_2 \ge \cdots \ge 0$ and $r = (\sum_j \lambda_j^2)^{1/2}$,

$$\frac{1}{2} r \sum_{j\ge 2} \lambda_j^2 \le \Phi(T) \le \frac{3}{2} r \sum_{j\ge 2} \lambda_j^2; \qquad \text{see (5.7).}$$

Lemma 5.5 expresses the first two noise derivatives of T_m as relative quadratic forms. Applying Lemma 5.4 then bounds the corresponding derivatives of Φ by Φ itself, without a spectral gap and with constants independent of the operator trace.

(5) **Passing to weak solutions and estimating exit times.** Regularizing the tensor kernels in the weak equation transfers these relative estimates to measure-valued solutions. Squared noise-field differences suppress the singular Hessian terms in the limit, including on atomic diagonals, and give balance identities throughout the weak solution class. In particular,

$$\mathrm{d}\mathcal{Z}_t = \mathcal{Z}_t \left(\alpha_t \, \mathrm{d}t + \sum_k \beta_{t,k} \, \mathrm{d}W_t^k \right),$$
$$|\alpha_t| \le C_Z \varepsilon^2, \qquad \sum_k |\beta_{t,k}|^2 \le V_Z \varepsilon^2.$$

Maximal and logarithmic martingale inequalities for the nonnegative supermartingale $e^{-C_Z\varepsilon^2 t}\mathcal{Z}(u_t)$ give Theorem 1.3(ii); combining the maximal bound with step (3) gives assertion (i). A stopped expectation argument gives the mean-exit bound in assertion (iii), while nonnegativity and continuity prove preservation of $\mathcal{P}$ for nonzero initial momentum with vanishing defect.

Section 2 introduces the weak solution class and develops positive particle dynamics and quadratic balances. Section 3 constructs solutions for general nonnegative momenta. Section 4 uses ordered characteristics to prove pathwise uniqueness and continuous dependence in the narrow topology. Section 5 develops the relative cubic-defect estimates and derives the exit bounds. Section 6 relates the stability estimates to exact peakon dynamics and particle computations, describing amplitude evolution, shape persistence and the cubic defect. Appendix A supplies the stochastic-flow and commutator calculations used in the pathwise theory.

2. Weak solutions, positive particles and quadratic balances

Let us begin with a weak formulation that accommodates atomic momentum and will also describe the particle approximations constructed below.

Let $\mathcal{M}_+(\mathbb{R})$ denote the space of nonnegative finite Radon measures on $\mathbb{R}$, and set

$$G(x)=\frac{1}{2}e^{-|x|},\quad G'(0)=0,\quad u_m=G*m,\quad M(m)=m(\mathbb{R}).$$

The assigned value of G' at the origin is the mean of its two one-sided limits. The space $\mathcal{M}_+(\mathbb{R})$ carries the narrow topology: convergence is tested against all bounded continuous functions and includes convergence of total mass.

For $f\in C_c^\infty(\mathbb{R})$, set

$$B_kf=\xi_kf'-\xi_k'f,\quad B_k^2f=\xi_k^2f''-\xi_k\xi_k'f'+\big((\xi_k')^2-\xi_k\xi_k''\big)f,$$

$$F_f(m)=\iint_{\mathbb{R}^2}K_f(x,y)m(\mathrm{d}x)m(\mathrm{d}y), \tag{2.1}$$

$$K_f(x,y)=\frac{1}{2}G(x-y)\big(f'(x)+f'(y)\big)-\frac{1}{2}G'(x-y)\big(f(x)-f(y)\big). \tag{2.2}$$

The last product in (2.2) extends continuously across the diagonal. Thus (2.1) defines the nonlinear drift for atomic momentum without multiplying an atom by a discontinuous derivative.

Definition 2.1. A global weak martingale solution with nonnegative momentum and initial condition $m_0\in\mathcal{M}_+(\mathbb{R})$ consists of a filtered probability space satisfying the usual conditions, an N-dimensional Brownian motion W, and an adapted process $m\in C([0,\infty);\mathcal{M}_+(\mathbb{R}))$ such that $m(0)=m_0$ and, for each $f\in C_c^\infty(\mathbb{R})\cup\{1\}$,

$$\langle f,m_t\rangle=\langle f,m_0\rangle+\int_0^t F_f(m_s)\,\mathrm{d}s+\frac{\varepsilon^2}{2}\sum_{k=1}^N\int_0^t\langle B_k^2f,m_s\rangle\,\mathrm{d}s+\varepsilon\sum_{k=1}^N\int_0^t\langle B_kf,m_s\rangle\,\mathrm{d}W_s^k. \tag{2.3}$$

The identity holds almost surely for all $t\ge0$ for each fixed test function. The process must also satisfy $\mathbb{E}\sup_{t\le T}M(m_t)^p<\infty$ for every $T<\infty$ and $p\ge2$.

Within this solution class, consider the nonnegative quadratic functional

$$D(m)=\frac{1}{2}M(m)^2-E(m)=\frac{1}{2}\iint_{\mathbb{R}^2}K(x,y)m(\mathrm{d}x)m(\mathrm{d}y),$$

$$K(x,y)=1-e^{-|x-y|}.$$

Write $D_t=D(m_t)$ and $D_0=D(m_0)$; the definition also gives $0\le D(m)\le M(m)^2/2$. For one vector field ξ, introduce the pair operator

$$\mathcal{T}_\xi=\xi(x)\partial_x+\xi(y)\partial_y-\xi'(x)-\xi'(y)$$

and the quadratic functionals

$$A_k(m) = \frac{1}{2}\iint_{\mathbb{R}^2} \mathcal{T}_{\xi_k} K \,\mathrm{d}m\,\mathrm{d}m, \quad C_k(m) = \frac{1}{2}\iint_{\mathbb{R}^2} \mathcal{T}_{\xi_k}^2 K \,\mathrm{d}m\,\mathrm{d}m. \tag{2.4}$$

The kernels in (2.4) are defined by their continuous extensions from $x \neq y$; explicit formulas appear in Lemma 2.5.

Write

$$L_{j,k} = \|\xi_k^{(j)}\|_\infty, \quad c_k = 10L_{1,k}^2 + 3L_{0,k}L_{2,k}.$$

Theorem 2.2 (Existence and stochastic balance for D)**.** *For every deterministic $m_0 \in \mathcal{M}_+(\mathbb{R})$, equation (1.1) has a global weak martingale solution with nonnegative momentum obtained by positive atomic approximation. Moreover, for every solution in Definition 2.1, $u = G * m \in C([0,\infty); H^1(\mathbb{R}))$ almost surely, and*

$$D(m_t) = D(m_0) + \varepsilon \sum_k \int_0^t A_k(m_s)\,\mathrm{d}W_s^k + \frac{\varepsilon^2}{2}\sum_k \int_0^t C_k(m_s)\,\mathrm{d}s. \tag{2.5}$$

The coefficients satisfy

$$|A_k(m)| \le 3L_{1,k}D(m), \quad |C_k(m)| \le c_k D(m). \tag{2.6}$$

If $M(m_0) > 0$, then $M(m_t) > 0$ at every finite time almost surely. If $D(m_0) > 0$, then $D(m_t) > 0$ at every finite time almost surely; if $D(m_0) = 0$, the functional remains zero.

The approximation argument begins with continuity of the energy and the construction of global positive particle solutions.

Lemma 2.3 (Energy continuity and vanishing quadratic defect)**.** *The maps $m \mapsto G * m$ and $m \mapsto D(m)$ are continuous from $\mathcal{M}_+(\mathbb{R})$ with the narrow topology into H^1 and $\mathbb{R}$, respectively. If $m \neq 0$, then $D(m) = 0$ if and only if m is a single atom.*

Proof. The Bochner integral $G * m$ belongs to H^1, since all translates of G have the same finite H^1 norm and m is finite. The identity $\langle G(\cdot - x), G(\cdot - y)\rangle_{H^1} = G(x-y)$ proves (1.4). If $m_n \to m$ narrowly, convergence of product measures and the boundedness and continuity of $G(x-y)$ imply

$$\begin{aligned}\|G * (m_n - m)\|_{H^1}^2 = &\iint_{\mathbb{R}^2} G(x-y)m_n(\mathrm{d}x)m_n(\mathrm{d}y) + \iint_{\mathbb{R}^2} G(x-y)m(\mathrm{d}x)m(\mathrm{d}y)\\ &- 2\iint_{\mathbb{R}^2} G(x-y)m_n(\mathrm{d}x)m(\mathrm{d}y) \longrightarrow 0.\end{aligned}$$

Mass is continuous in the narrow topology, so D is continuous.

The kernel K is positive off the diagonal. Thus $D = 0$ means that the product of the normalized measure with itself is concentrated on the diagonal, which is equivalent to the normalized measure being a Dirac mass. □

The static identities connect the weak framework to finite-dimensional dynamics. Consider initially ordered particles with positive weights:

$$m_t^n = \sum_{i=1}^n p_i(t)\delta_{q_i(t)}, \quad p_i(0) > 0, \quad q_1(0) < \cdots < q_n(0).$$

The corresponding Stratonovich system is

$$\mathrm{d}q_i = u(q_i)\,\mathrm{d}t + \varepsilon\sum_k \xi_k(q_i) \circ \mathrm{d}W^k, \tag{2.7}$$

$$\mathrm{d}p_i = -p_i u_x(q_i)\,\mathrm{d}t - \varepsilon p_i \sum_k \xi_k'(q_i) \circ \mathrm{d}W^k, \tag{2.8}$$

where $u = \sum_i p_i G(\cdot - q_i)$ and $u_x(q_i) = \sum_j p_j G'(q_i - q_j)$ with $G'(0) = 0$. For a single particle, the peak height is $p_1/2$.

Proposition 2.4 (Global existence for positive particle solutions)**.** *System* (2.7)–(2.8) *has a global solution. The positions remain strictly ordered and all weights remain positive. Its empirical measure satisfies* (2.3)*. For each $p \geq 2$ and $T < \infty$,*

$$\mathbb{E} \sup_{t \leq T} M(m_t^n)^p \leq C_{p,\xi,\varepsilon,T} M(m_0^n)^p, \tag{2.9}$$

where the constant is independent of the number and positions of the particles.

Proof. The coefficients are locally Lipschitz on the region with positive weights and strictly ordered positions. Antisymmetry of G' gives $\sum_i p_i u_x(q_i) = 0$. Stratonovich-to-Itô conversion therefore gives

$$\mathrm{d}M = -\varepsilon \sum_k \langle \xi_k', m^n \rangle \, \mathrm{d}W^k + \frac{\varepsilon^2}{2} \sum_k \langle (\xi_k')^2 - \xi_k \xi_k'', m^n \rangle \, \mathrm{d}t. \tag{2.10}$$

The diffusion and drift are bounded by constant multiples of εM and $\varepsilon^2 M$, respectively. Up to the localizing stopping time, the logarithm of M has bounded drift and diffusion coefficients, so the mass cannot explode or vanish at a finite lifetime.

The following moment estimate will also be used below. Suppose that, up to a localizing stopping time, a nonnegative continuous semimartingale Y satisfies

$$\mathrm{d}Y = b \, \mathrm{d}t + \sigma \cdot \mathrm{d}W, \quad b \leq aY + cX, \quad |\sigma| \leq aY,$$

where Y_0 is deterministic, $X \geq 0$, and $a, c \geq 0$ are deterministic constants. For $p \geq 2$, Itô's formula and Young's inequality give

$$Y_t^p \leq Y_0^p + C \int_0^t (Y_s^p + X_s^p) \, \mathrm{d}s + L_t, \quad L_t = p \int_0^t Y_s^{p-1} \sigma_s \cdot \mathrm{d}W_s.$$

All processes in this calculation are stopped at the same localizing time. The Burkholder–Davis–Gundy inequality and Young's inequality imply

$$\mathbb{E} \sup_{r \leq t} |L_r| \leq C \mathbb{E} \left(\sup_{r \leq t} Y_r^p \int_0^t Y_s^p \, \mathrm{d}s \right)^{1/2} \leq \frac{1}{2} \mathbb{E} \sup_{r \leq t} Y_r^p + C \int_0^t \mathbb{E} Y_s^p \, \mathrm{d}s.$$

Consequently, Gronwall's inequality yields

$$\mathbb{E} \sup_{t \leq T} Y_t^p \leq C_{p,a,c,T} \left(Y_0^p + \mathbb{E} \int_0^T X_s^p \, \mathrm{d}s \right).$$

Fatou's lemma removes the localization whenever the stopped processes converge on $[0, T]$. With $Y = M$ and $X = 0$, this proves the mass estimate. For the weighted mass, take $Y = \langle V, m \rangle$ and $X = M^2$; the mass estimate at exponent $2p$ then gives the claimed weighted moment bound.

For the individual weights, the logarithmic equations are

$$\mathrm{d} \log p_i = -u_x(q_i) \, \mathrm{d}t - \varepsilon \sum_k \xi_k'(q_i) \, \mathrm{d}W^k - \frac{\varepsilon^2}{2} \sum_k \xi_k(q_i) \xi_k''(q_i) \, \mathrm{d}t.$$

Since $|u|, |u_x| \leq M/2$, each weight remains positive and finite on every finite time interval. The position equations have integrable drift and bounded diffusion coefficients on finite time intervals.

For a positive gap $z = q_j - q_i$, its Itô diffusion coefficients are $\varepsilon(\xi_k(q_j) - \xi_k(q_i))$. Its drift is

$$u(q_j) - u(q_i) + \frac{\varepsilon^2}{2} \sum_k \big((\xi_k \xi_k')(q_j) - (\xi_k \xi_k')(q_i) \big).$$

After division by z, the diffusion coefficients are bounded and the drift is bounded by $M/2 + C\varepsilon^2$. Itô's formula for $\log z$, stopped before a collision, consequently precludes a finite-time collision. These estimates allow us to remove the localization and continue the solution globally.

The Stratonovich chain rule for $\sum_i p_i f(q_i)$ gives $\langle uf' - u_x f, m^n \rangle$ as the deterministic drift and $\varepsilon \langle B_k f, m^n \rangle$ as the noise coefficient. Symmetrizing the deterministic double sum gives (2.1), and converting the noise to Itô form gives $B_k^2/2$. Thus (2.3) follows. □

To derive the quadratic balance, we next establish a uniform estimate for the kernels that define its coefficients.

Lemma 2.5 (Bounds for the kernels defining A_k and C_k)**.** *For one $\xi \in C_b^3$, let $L_j = \|\xi^{(j)}\|_\infty$. For $x \neq y$, put*

$$r = |x-y|, \quad s = \operatorname{sgn}(x-y), \quad e = e^{-r}, \quad a = \xi(x) - \xi(y), \quad b = \xi'(x) + \xi'(y),$$
$$w = \xi(x)\xi'(x) - \xi(y)\xi'(y), \quad h = \xi(x)\xi''(x) + \xi(y)\xi''(y).$$

Then

$$\mathcal{T}_\xi K = sae - bK, \tag{2.11}$$
$$\mathcal{T}_\xi^2 K = e\{s(w-2ba) - a^2\} + (b^2 - h)K. \tag{2.12}$$

Both kernels extend continuously by zero to $x = y$, and

$$|\mathcal{T}_\xi K| \le 3L_1 K,$$
$$|\mathcal{T}_\xi^2 K| \le (10L_1^2 + 3L_0L_2)K.$$

Proof. Let $V = \xi(x)\partial_x + \xi(y)\partial_y$. Away from the diagonal, $VK = sae$, $Va = w$, $Vb = h$, and $Ve = -sae$. Since $\mathcal{T}_\xi = V - b$, differentiation gives (2.11)–(2.12). The inequalities

$$re^{-r} \le 1 - e^{-r}, \quad r^2 e^{-r} \le 1 - e^{-r} \quad (r \ge 0)$$

are elementary. For the second inequality, $e^r - 1 - r^2$ is increasing from zero because $e^r \ge 2r$. The following bounds will be used:

$$|a| \le L_1 r, \quad |b| \le 2L_1, \quad |w| \le (L_1^2 + L_0L_2)r, \quad |h| \le 2L_0L_2.$$

The five terms in (2.12) are bounded in absolute value, respectively, by

$$(L_1^2 + L_0L_2)K, \quad 4L_1^2K, \quad L_1^2K, \quad 4L_1^2K, \quad 2L_0L_2K.$$

This proves the stated bounds and continuity at the diagonal. □

3. Global weak solutions and stochastic balance identities

We construct a weak solution by positive particle approximation. A coercive weight gives compact containment without a spatial moment assumption, and joint martingale identities identify the limiting equation. The quadratic balance is then derived directly for every solution in Definition 2.1, completing Theorem 2.2. If $m_0 = 0$, take $m_t = 0$ for every t. For the rest of the construction, assume $M(m_0) > 0$.

3.1. A coercive weight without a spatial moment assumption

Lemma 3.1. *For every $m_0 \in \mathcal{M}_+(\mathbb{R})$ there exists $V \in C^\infty(\mathbb{R})$ such that $V \ge 1$, $V(x) \to \infty$ as $|x| \to \infty$, $\|V^{(j)}\|_\infty < \infty$ for every $j \ge 1$, and $\langle V, m_0\rangle < \infty$. There are finite positive atomic measures m_0^n that have the same mass as m_0, converge narrowly to m_0, and satisfy $\sup_n \langle V, m_0^n\rangle < \infty$.*

Proof. Choose a smooth even $\chi : \mathbb{R} \to [0,1]$ which is zero on $[-1,1]$ and one outside $(-2,2)$. Choose an increasing sequence $R_j \ge 2^j$ such that $m_0(\{|x| > R_j\}) \le 2^{-j}$, and set

$$V(x) = 1 + \sum_{j=1}^{\infty} \chi(x/R_j).$$

The sum is locally finite, every positive-order derivative is bounded, and V is coercive. Also $\langle V, m_0\rangle \le M(m_0) + 1$. To construct m_0^n, move the mass outside $[-n,n]$ to the origin and quantize the remaining mass on a mesh of size at most $1/n$. Then

$$\langle V, m_0^n\rangle \le \langle V, m_0\rangle + \|V'\|_\infty M(m_0)/n.$$

Mass is preserved, and the approximation converges narrowly. Atoms at the same point may be merged. □

Let m^n be the global particle solutions with these initial conditions. Put $M_t^n = M(m_t^n)$ and $Z_t^n = \langle V, m_t^n \rangle$. For $f = V$, formula (2.3) follows directly from the finite particle sum, so it remains valid for the unbounded test function V. The deterministic drift satisfies

$$|F_V(m)| \le C\|V'\|_\infty M(m)^2.$$

Indeed, we bound the first term in K_V using V', and the second using $|V(x) - V(y)| \le \|V'\|_\infty |x - y|$ and the boundedness of $|rG'(r)|$. Since $V \ge 1$ and its first two derivatives are bounded,

$$|B_k V| + |B_k^2 V| \le C_{V,\xi} V.$$

Consequently, the drift of Z^n is bounded in absolute value by $C(M^n)^2 + C\varepsilon^2 Z^n$, and its diffusion vector in norm by $C\varepsilon Z^n$. For $p \ge 2$, localization, Itô's formula and the BDG inequality, followed by Young's inequality to absorb the supremum term, give

$$\sup_n \mathbb{E} \sup_{t \le T} (Z_t^n)^p < \infty \quad \text{for every } T < \infty. \tag{3.1}$$

The left-hand side is bounded by a constant depending on $p, V, \xi, \varepsilon, T$, multiplied by $1 + \sup_n (Z_0^n)^p + \sup_n \mathbb{E} \sup_{t \le T} (M_t^n)^{2p}$. The last term is finite by (2.9).

3.2. Tightness of continuous measure paths

The space $\mathcal{M}_+(\mathbb{R})$ with its narrow topology is Polish. For $R > 0$, the set

$$\mathcal{K}_R = \{m \in \mathcal{M}_+(\mathbb{R}) : \langle V, m \rangle \le R\}$$

is compact: masses are bounded, coercivity of V gives uniform tightness, and lower semicontinuity of the V integral gives closedness. Estimate (3.1) therefore implies

$$\lim_{R \to \infty} \sup_n \mathbb{P}(m_t^n \notin \mathcal{K}_R \text{ for some } t \le T) = 0. \tag{3.2}$$

For $f \in C_c^\infty(\mathbb{R}) \cup \{1\}$, the functions K_f, $B_k f$ and $B_k^2 f$ are bounded and continuous. In particular,

$$|F_f(m)| \le C_f M^2, \quad |\langle B_k f, m \rangle| + |\langle B_k^2 f, m \rangle| \le C_f M. \tag{3.3}$$

The particle weak equation, the BDG inequality and the mass moment bounds give

$$\sup_n \mathbb{E} |\langle f, m_t^n - m_s^n \rangle|^4 \le C_{f,T} |t - s|^2, \quad 0 \le s < t \le T.$$

Choose a countable separating family consisting of 1 and smooth compactly supported functions dense in $C_0(\mathbb{R})$. It can also be chosen dense in the C^2 norm on each fixed compact support when needed below. On each $\mathcal{K}_R$, the coordinate topology is the narrow topology. The coordinate Kolmogorov criterion, (3.2), and the Arzelà–Ascoli theorem show tightness in $C([0,T]; \mathcal{M}_+(\mathbb{R}))$. For completeness, one can impose a compact coordinate-path constraint for the jth test with failure probability at most $\delta 2^{-j}$, intersect these constraints with (3.2), and use a diagonal sequence. The resulting set is relatively compact in the measure-path space.

The joint laws of (m^n, W) are therefore tight in the product of $C([0,T]; \mathcal{M}_+(\mathbb{R}))$ and $C([0,T]; \mathbb{R}^N)$. Applying this argument for every integer T gives tightness in the corresponding locally uniform path spaces on $[0, \infty)$. Prokhorov's theorem and Skorokhod's representation theorem [37, 39, 19] then give a new probability space and a subsequence such that

$$(m^n, W^n) \longrightarrow (m, W) \quad \text{almost surely, locally uniformly in time.} \tag{3.4}$$

We retain the same notation for the represented processes; the limit has narrowly continuous paths of nonnegative measures. Evaluation at $t = 0$ in (3.4), together with $m_0^n \to m_0$ narrowly, gives the initial condition $m(0) = m_0$. Both the mass moment estimates and the weighted mass estimate pass to the limit by lower semicontinuity.

3.3. Identification of the weak equation

To identify the stochastic integrals in (3.4), we pass to the limit in the joint martingale identities for the linear tests and Brownian coordinates. These identities also show that the limiting noise is Brownian in the joint filtration. For a test function f in the countable family fixed above, set

$$a_f(m) = F_f(m) + \frac{\varepsilon^2}{2}\sum_k \langle B_k^2 f, m\rangle, \quad b_{f,k}(m) = \varepsilon\langle B_k f, m\rangle,$$

and define

$$N_f^n(t) = \langle f, m_t^n\rangle - \langle f, m_0^n\rangle - \int_0^t a_f(m_s^n)\,\mathrm{d}s.$$

Narrow convergence of finite positive measures implies convergence of their product measures. Hence a_f and $b_{f,k}$ are continuous. Let i, j range over the chosen countable family of test functions. For particle solutions, the following processes are martingales:

$$N_i^n, \quad W^{n,k}, \tag{3.5}$$

$$N_i^n N_j^n - \int_0^{\cdot}\sum_k b_{i,k}(m_s^n)b_{j,k}(m_s^n)\,\mathrm{d}s,$$

$$N_i^n W^{n,k} - \int_0^{\cdot} b_{i,k}(m_s^n)\,\mathrm{d}s, \quad W^{n,k}W^{n,l} - \delta_{kl}t. \tag{3.6}$$

All the displayed random variables are uniformly integrable on each bounded time interval. Indeed, the coefficients of each linear test are bounded by fixed polynomials in M, and (2.9) provides every required moment. Brownian moment bounds control the factors containing W.

Multiply the increments of the processes in (3.5)–(3.6) by bounded continuous cylinder functions of the joint past (m^n, W^n). Continuity, (3.4) and uniform integrability allow passage to the limit. A monotone-class argument extends the limiting identities to all bounded measurable functions of the past. Thus W is a continuous martingale in the joint canonical filtration with $[W^k, W^l]_t = \delta_{kl}t$. By Lévy's characterization, it is a Brownian motion in that filtration.

Moreover, the limiting martingales N_i satisfy

$$[N_i, N_j]_t = \int_0^t \sum_k b_{i,k}(m_s)b_{j,k}(m_s)\,\mathrm{d}s, \quad [N_i, W^k]_t = \int_0^t b_{i,k}(m_s)\,\mathrm{d}s.$$

It follows that the continuous local martingale

$$N_i - \sum_k \int_0^{\cdot} b_{i,k}(m_s)\,\mathrm{d}W_s^k$$

has zero quadratic variation and hence vanishes. This proves (2.3) for the countable family, with its stochastic integrals represented by the limiting Brownian motions. The usual augmentation of the filtration preserves these statements. Approximation in C^2 on compact supports, the bounds (3.3), and the Itô isometry extend the weak equation to every test function in C_c^2. In particular, $B_k f \in C_c^2$ for $f \in C_c^\infty$ because $\xi_k \in C_b^3$. Testing with $B_k f$ identifies $[\langle B_k f, m\rangle, W^k]_t = \varepsilon\int_0^t \langle B_k^2 f, m_s\rangle\,\mathrm{d}s$, which recovers the Stratonovich formulation (1.1). The test 1 was retained throughout, so the mass equation (2.10) holds in the limit. Thus the limit is a solution in the sense of Definition 2.1.

Finally, Lemma 2.3 gives continuity of $m \mapsto G * m$ into H^1. On each finite time interval, the union of the ranges of a uniformly convergent sequence of continuous measure paths and its limit is relatively compact. The map is uniformly continuous on the compact closure of this union, so (3.4) also yields

$$\sup_{t\le T}\|G * m_t^n - G * m_t\|_{H^1} \longrightarrow 0 \quad \text{almost surely.}$$

The same continuity lemma gives continuous H^1 paths for every solution in Definition 2.1.

3.4. The quadratic balance for every weak solution

Proposition 3.2. *Every solution in Definition 2.1 satisfies (2.5) on its original filtered probability space and with its original Brownian motions.*

Proof. Fix such a solution, and write

$$u = G * m, \quad v = G' * m, \quad |u|, |v| \le M/2.$$

The value $G'(0) = 0$ is retained. Although v can be discontinuous, it is bounded and Borel measurable, and the symmetrized drift can be written as

$$F_f(m) = \int_{\mathbb{R}} (uf' - vf)\, \mathrm{d}m. \tag{3.7}$$

The integrals are absolutely convergent.

Step 1: tensor Itô formula. For a smooth bounded kernel $h(x, y)$ with bounded derivatives through order two, put

$$Q_h(m) = \iint_{\mathbb{R}^2} h(x, y) m(\mathrm{d}x) m(\mathrm{d}y),$$

$$\mathcal{L}_m h = u(x)\partial_x h + u(y)\partial_y h - (v(x) + v(y))h.$$

Then

$$\mathrm{d}Q_h(m_t) = Q_{\mathcal{L}_{m_t} h}(m_t)\, \mathrm{d}t + \varepsilon \sum_k Q_{\mathcal{T}_{\xi_k} h}(m_t)\, \mathrm{d}W_t^k + \frac{\varepsilon^2}{2} \sum_k Q_{\mathcal{T}_{\xi_k}^2 h}(m_t)\, \mathrm{d}t. \tag{3.8}$$

To justify this identity from the weak formulation, first take $h(x, y) = f(x)g(y)$ with $f, g \in C_c^\infty$ and use the ordinary Itô product formula. The cross variation of the two factors is $\varepsilon^2 \sum_k \langle B_k f, m\rangle \langle B_k g, m\rangle\, \mathrm{d}t$, which gives the mixed part of $\frac{\varepsilon^2}{2}(B_k^x + B_k^y)^2 h$. Equation (3.7) gives the deterministic term. Finite sums of these tensors approximate any smooth compactly supported kernel in the uniform C^2 norm, for example by Fourier approximation on a rectangle followed by separate cutoffs. The deterministic drift errors are bounded by $CM^3\|h - h_n\|_{C^1}$, the diffusion errors by $CM^2\|h - h_n\|_{C^1}$, and the Itô correction errors by $CM^2\|h - h_n\|_{C^2}$. The mass moment bounds and the BDG inequality justify passage to the limit.

For a kernel as above that is not compactly supported, apply this result to $h_R(x, y) = \chi(x/R)\chi(y/R)h(x, y)$, where χ is a smooth compactly supported cutoff equal to one near zero. For $R \ge 1$, the derivatives through order two are uniformly bounded. The third and fourth mass moments allow passage to the limit in the time and stochastic integrals by dominated convergence. The range of a narrowly continuous measure path on $[0, T]$ is compact, hence uniformly tight. This gives uniform-in-time convergence of $Q_{h_R}(m_t)$ to $Q_h(m_t)$ and proves (3.8). No spatial moment condition is used.

Step 2: deterministic energy drift. Choose an even nonnegative $\rho \in C_c^\infty((-1, 1))$ with integral one, and set

$$\rho_\delta(r) = \delta^{-1}\rho(r/\delta), \quad G_\delta = \rho_\delta * G, \quad u_\delta = G_\delta * m, \quad E_\delta = Q_{G_\delta(x-y)}(m).$$

These kernels satisfy

$$G_\delta'' = G_\delta - \rho_\delta, \quad \|G_\delta - G\|_\infty \le C\delta, \quad \|G_\delta'\|_\infty \le \tfrac{1}{2}, \quad G_\delta'(r) \longrightarrow G'(r) \quad (r \in \mathbb{R}). \tag{3.9}$$

The last convergence also holds at $r = 0$, because both derivatives are zero there. Formula (3.8) gives the deterministic drift of E_δ as

$$R_\delta(m) = 2 \int_{\mathbb{R}} \left[u(G_\delta' * m) - vu_\delta\right]\, \mathrm{d}m.$$

For each fixed finite positive measure, bounded convergence, first in the convolution and then in the outer integral, gives $R_\delta(m) \to 0$. Also $|R_\delta(m)| \le M^3$. Thus

$$\mathbb{E} \int_0^T |R_\delta(m_t)|\, \mathrm{d}t \longrightarrow 0. \tag{3.10}$$

The mean-derivative convention also makes the argument valid for atoms.

Step 3: the singular noise correction. For one field ξ, use a, b, w, h from Lemma 2.5. For any smooth scalar function $g(r)$, $r = x - y$, differentiation yields

$$\mathcal{T}_\xi g = ag' - bg, \quad \mathcal{T}_\xi^2 g = a^2 g'' + (w - 2ba)g' + (b^2 - h)g.$$

In particular,

$$\mathcal{T}_\xi^2 G_\delta = a^2 G_\delta - a^2 \rho_\delta + (w - 2ba)G'_\delta + (b^2 - h)G_\delta.$$

The only singular term satisfies

$$\sup_{x,y} |a^2 \rho_\delta(x - y)| \le L_1^2 \delta \sup_z z^2 \rho(z) \longrightarrow 0.$$

All remaining terms converge uniformly. Indeed, when $|x - y| > \delta$, G' is Lipschitz on the half-line on which the convolution is evaluated, so $|G'_\delta - G'| \le C\delta$. When $|x-y| \le \delta$, this difference is bounded by one, while $|a| \le L_1|x-y|$ and $|w-2ba| \le (5L_1^2+L_0L_2)|x-y|$. Both coefficients are also globally bounded. Together with (3.9), these observations show, for $K_\delta(x, y) = 1 - 2G_\delta(x - y)$, that

$$\|\mathcal{T}_\xi K_\delta - \mathcal{T}_\xi K\|_\infty + \|\mathcal{T}_\xi^2 K_\delta - \mathcal{T}_\xi^2 K\|_\infty \le C_{\rho,\xi}\delta. \tag{3.11}$$

The limits are the continuous kernels of Lemma 2.5, with no diagonal measure or local-time correction.

Step 4: removal of regularization. Let $D_\delta = M^2/2 - E_\delta$ and define $A_{k,\delta}, C_{k,\delta}$ by replacing K with K_δ in (2.4). The mass has no deterministic CH drift, since $\int_{\mathbb{R}} v \, \mathrm{d}m = 0$ by oddness of G'. Equation (3.8) therefore gives

$$\mathrm{d}D_\delta = -R_\delta \, \mathrm{d}t + \varepsilon \sum_k A_{k,\delta} \, \mathrm{d}W^k + \frac{\varepsilon^2}{2} \sum_k C_{k,\delta} \, \mathrm{d}t.$$

The bounds above imply

$$|D_\delta - D| \le C\delta M^2, \quad |A_{k,\delta} - A_k| + |C_{k,\delta} - C_k| \le C_{\rho,\xi}\delta M^2.$$

Consequently, the regularized functional converges in L^2 with respect to the uniform norm on $[0, T]$. The Itô isometry and the BDG inequality give convergence of the stochastic integrals, while the correction drifts converge in L^1 and the remaining drift vanishes by (3.10). Passing to the limit therefore proves (2.5) as an identity of continuous semimartingales on the given stochastic basis. □

The regularization also gives an explicit energy balance. For $j = 1, 2$, define

$$\mathcal{E}_k^{(j)}(m) = \iint_{\mathbb{R}^2} \mathcal{T}_{\xi_k}^j G(x - y) \, m(\mathrm{d}x)m(\mathrm{d}y),$$

where the kernels are defined by their continuous extensions across $x = y$. For every solution in Definition 2.1,

$$\mathrm{d}E(m_t) = \varepsilon \sum_k \mathcal{E}_k^{(1)}(m_t) \, \mathrm{d}W_t^k + \frac{\varepsilon^2}{2} \sum_k \mathcal{E}_k^{(2)}(m_t) \, \mathrm{d}t. \tag{3.12}$$

Indeed, apply (3.8) directly to $G_\delta(x-y)$. Its deterministic drift is R_δ, which tends to zero by (3.10). The first and second noise kernels converge uniformly by the estimates used in (3.11), since $G_\delta = (1 - K_\delta)/2$. The mass moment bounds and the BDG inequality justify passage to the limit in the stochastic integrals. These coefficients are the first and second ordinary derivatives of E along the corresponding deterministic coadjoint flow.

The bounds (2.6) now follow from Lemma 2.5 for every nonnegative momentum measure. The mass equation has the form $\mathrm{d}M = M\alpha_t \, \mathrm{d}t + M\beta_t \cdot \mathrm{d}W$, with bounded adapted coefficients obtained by dividing its coefficients by M on $\{M > 0\}$ and setting the ratios to zero at zero. The stochastic exponential representation of this linear equation proves strict positivity at every finite time when $M_0 > 0$; if $M_0 = 0$, the solution is identically zero. By (2.6), the same argument applies to (2.5): its coefficients divided by D on $\{D > 0\}$ are bounded and vanish at $D = 0$. Thus $D_t > 0$ at every finite time when $D_0 > 0$, whereas D remains zero when $D_0 = 0$. These arguments take place on the original stochastic basis of each weak solution. Together with the preceding construction and continuity statement, they complete Theorem 2.2.

4. Pathwise well-posedness in Lagrangian coordinates

The proof of Theorem 1.1 starts by expressing every weak solution in terms of weighted characteristics. Comparing these characteristics using the order of their initial labels gives pathwise uniqueness, after which positive particle approximations on the prescribed Brownian space yield a probabilistically strong solution and continuous dependence.

Let ϕ_t be the noise-only flow of Lemma A.1, and write $J_t = \partial_x \phi_t$, $H_t = \partial_x^2 \phi_t$, and $a = \varepsilon^2 \sum_k \xi_k \xi_k'/2$ for its Itô drift. Appendix A proves the flow regularity under C_b^3 noise, the random-test identity, and the technical characteristic estimates used below. Only local bounds on H are needed.

4.1. Characteristic representation and uniform bounds

Define

$$\nu_t = (\phi_t^{-1})_{\#}\big((J_t \circ \phi_t^{-1}) m_t\big), \quad m_t = (\phi_t)_{\#}(J_t^{-1} \nu_t).$$

The global Jacobian bound ensures that this is a path of nonnegative finite measures. Narrow continuity follows from joint local continuity of the maps and weights, their global bounds on finite intervals, bounded displacement of ϕ, and uniform tightness of a continuous measure path; no global spatial Lipschitz bound for J is needed.

The random-test identity (A.7), proved in Appendix A.2, gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle f, \nu_t \rangle = \langle v_t f' + b_t f, \nu_t \rangle,$$

where

$$v_t(x) = \frac{u_t(\phi_t(x))}{J_t(x)}, \quad b_t(x) = -\bar{u}_{t,x}(\phi_t(x)) + u_t(\phi_t(x)) \frac{H_t(x)}{J_t(x)^2}.$$

The field v is continuous and bounded on $[0,T] \times \mathbb{R}$ for every finite T, and locally Lipschitz in space uniformly on compact time intervals. The reaction coefficient b is Borel measurable and bounded on compact space–time sets, with its values at atoms specified by $G'(0) = 0$. This local boundedness of b is sufficient for the argument.

Apply Lemma A.3 from Appendix A.3 pathwise. Let $Q_t = Q_{0,t}$ be the characteristic flow of v, and set

$$r_t(x) = \exp\left(\int_0^t b_s(Q_s(x))\, \mathrm{d}s \right), \quad q_t(x) = \phi_t(Q_t(x)), \quad p_t(x) = r_t(x)/J_t(Q_t(x)).$$

This gives (1.2), positivity of p, strict order, and a uniform displacement bound on each finite time interval. The fields admit versions that are jointly measurable in the sample point, time and initial label: the continuous adapted field v has a causal local Picard flow, and the exponent defining r is the time integral of a progressively measurable Borel field.

To recover the stochastic equations for these characteristics, consider the following composition formula. Suppose that, locally in space, a continuous semimartingale field satisfies

$$F_t(x) = F_0(x) + \int_0^t A_s(x)\, \mathrm{d}s + \sum_k \int_0^t C_{k,s}(x)\, \mathrm{d}W_s^k,$$

where F has a jointly continuous first spatial derivative, A is continuous in space, and the C_k are locally uniformly Lipschitz in space after localization. For an adapted absolutely continuous path Q with locally bounded derivative, one has

$$\mathrm{d}F_t(Q_t) = \big(A_t(Q_t) + \partial_x F_t(Q_t) \dot{Q}_t\big)\, \mathrm{d}t + \sum_k C_{k,t}(Q_t)\, \mathrm{d}W_t^k.$$

To verify this formula, split each increment into a time increment evaluated at the left-endpoint spatial argument and a spatial increment at the right time endpoint. The time-increment sums converge by dominated convergence and the Itô isometry. The spatial sums converge to the finite-variation integral by the fundamental theorem of calculus and uniform continuity of $\partial_x F$ on the localized compact set. There is no spatial second-order term because Q has zero quadratic variation. Apply this formula first to $F = \phi$ and then to $F = \log J$. For the latter field,

$$A = \frac{\varepsilon^2}{2} \sum_k (\xi_k \xi_k'')(\phi), \quad C_k = \varepsilon \xi_k'(\phi), \quad \partial_x F = H/J.$$

These coefficients satisfy the stated local conditions under C_b^3.

Applying this composition formula to ϕ gives

$$\mathrm{d}q = (u(q) + a(q))\,\mathrm{d}t + \varepsilon \sum_k \xi_k(q)\,\mathrm{d}W^k.$$

For $\log J$, the fixed-label Itô drift is $\varepsilon^2 \sum_k (\xi_k \xi_k'')(\phi)/2$. Composition contributes $(H/J)(Q)v(Q)\,\mathrm{d}t$. This cancels the corresponding term in $\mathrm{d}\log r$, giving

$$\mathrm{d}\log p = -\bar{u}_x(q)\,\mathrm{d}t - \varepsilon \sum_k \xi_k'(q)\,\mathrm{d}W^k - \frac{\varepsilon^2}{2} \sum_k (\xi_k \xi_k'')(q)\,\mathrm{d}t.$$

Exponentiation gives (1.3) and its Itô correction. These identities hold for each fixed label almost surely, and hence for m_0-almost every label simultaneously by Fubini, with versions continuous in time.

For the comparison of two solutions, the following lemma supplies uniform bounds on the characteristics.

Lemma 4.1. *For the representation above and every $T < \infty$, almost surely,*

$$\sup_{t \le T} \|q_t - \mathrm{id}\|_\infty < \infty, \quad \left\| \sup_{t \le T} (p_t + p_t^{-1}) \right\|_{L^\infty(m_0)} < \infty.$$

More precisely, fix $1/3 < \alpha < \beta < 1/2$ and a β-Hölder Brownian Stratonovich rough-path version, and let $\mathcal{R}_t$ be its running α rough-path seminorm. Before

$$\rho_R = \inf\{t \ge 0 : M_t + 1 + \mathcal{R}_t \ge R\},$$

the displacement and the essential upper bounds of p and p^{-1} on $[0,T]$ are bounded by deterministic constants depending on R, T, ε and the noise fields. The stopping times ρ_R tend to infinity almost surely.

The proof, including the identification of rough and Stratonovich integrals and the stopping-time construction, is given in Appendix A.4.

4.2. Pathwise uniqueness in Lagrangian coordinates

Consider two weak solutions with the same m_0 and W and the representations (q^1, p^1) and (q^2, p^2) above. Ordering gives the identity

$$G'(q_t^i(x) - q_t^i(y)) = -\tfrac{1}{2}\,\mathrm{sgn}(x-y) e^{-|q_t^i(x) - q_t^i(y)|}.$$

The sign is the same fixed function of the initial labels for both solutions, and is zero when $x = y$. Each initial atom corresponds to one spatial label and is not split into distinct ordered labels.

Write

$$\begin{aligned} U^i(x) &= \tfrac{1}{2} \int_{\mathbb{R}} e^{-|q^i(x) - q^i(y)|} p^i(y) m_0(\mathrm{d}y), \\ V^i(x) &= -\tfrac{1}{2} \int_{\mathbb{R}} \mathrm{sgn}(x-y) e^{-|q^i(x) - q^i(y)|} p^i(y) m_0(\mathrm{d}y), \\ h_k &= (\xi_k')^2 - \xi_k \xi_k''. \end{aligned}$$

The characteristic Itô equations are

$$\begin{aligned} \mathrm{d}q^i(x) &= \left[U^i(x) + \frac{\varepsilon^2}{2} \sum_k (\xi_k \xi_k')(q^i(x)) \right] \mathrm{d}t + \varepsilon \sum_k \xi_k(q^i(x))\,\mathrm{d}W^k, \\ \mathrm{d}p^i(x) &= \left[-p^i(x) V^i(x) + \frac{\varepsilon^2}{2} p^i(x) \sum_k h_k(q^i(x)) \right] \mathrm{d}t - \varepsilon p^i(x) \sum_k \xi_k'(q^i(x))\,\mathrm{d}W^k. \end{aligned} \tag{4.1}$$

Fix T. Let $\mathcal{R}_t$ denote the running α rough-path seminorm of the Brownian Stratonovich lift and put

$$\sigma_R = \inf\{t \geq 0 : M_t^1 + M_t^2 + 1 + \mathcal{R}_t \geq R\}.$$

These are stopping times and tend to infinity almost surely. Lemma 4.1 gives deterministic essential bounds on p^i and $(p^i)^{-1}$ in the initial label, with constants depending on R, T, before σ_R. The same localization bounds the displacements, without using a global bound on the second derivative of the noise flow. For $\Delta q = q^1 - q^2$ and $\Delta p = p^1 - p^2$, the Lipschitz property of $z \mapsto e^{-|z|}$ and the fixed sign give

$$\|U^1 - U^2\|_{L^2(m_0)} + \|V^1 - V^2\|_{L^2(m_0)} \leq C_{R,T,m_0}\big(\|\Delta q\|_{L^2(m_0)} + \|\Delta p\|_{L^2(m_0)}\big). \tag{4.2}$$

Indeed, if $|p^i| \leq P$, each of the two differences is bounded by

$$\frac{1}{2}\int_{\mathbb{R}} |\Delta p(y)| m_0(\mathrm{d}y) + \frac{P}{2}\int_{\mathbb{R}} \big(|\Delta q(x)| + |\Delta q(y)|\big) m_0(\mathrm{d}y).$$

The Cauchy–Schwarz inequality proves (4.2). Bounded weights and $|V^i| \leq M^i/2$ also control the difference of $p^i V^i$. The remaining Itô drift and noise coefficients in (4.1) satisfy the same local L^2 Lipschitz estimate. In particular, $h_k' = \xi_k' \xi_k'' - \xi_k \xi_k'''$ is bounded under C_b^3.

Set

$$Z_t = \int_{\mathbb{R}} \big(|\Delta q_t(x)|^2 + |\Delta p_t(x)|^2\big) m_0(\mathrm{d}x).$$

The individual Hilbert-space states may be written as $(q^i - \mathrm{id}, p^i - 1)$, whose initial values are zero. Thus no spatial moment of m_0 is needed to apply Itô's formula to their difference. Let $\mathsf{H} = L^2(m_0) \times L^2(m_0)$ and let ΔX denote $(\Delta q, \Delta p)$. Write ΔA and ΔS_k for the differences of the characteristic Itô drift and diffusion coefficients. The preceding estimates give, before σ_R,

$$\|\Delta A\|_{\mathsf{H}} + \sum_k \|\Delta S_k\|_{\mathsf{H}} \leq C_{R,T,m_0} \|\Delta X\|_{\mathsf{H}}.$$

Itô's formula therefore yields

$$Z_{t\wedge\sigma_R} = \int_0^t \mathbf{1}_{\{s<\sigma_R\}} \left(2\langle \Delta X_s, \Delta A_s\rangle_{\mathsf{H}} + \sum_k \|\Delta S_{k,s}\|_{\mathsf{H}}^2 \right) \mathrm{d}s + 2\sum_k \int_0^{t\wedge\sigma_R} \langle \Delta X_s, \Delta S_{k,s}\rangle_{\mathsf{H}} \, \mathrm{d}W_s^k.$$

The bounds on the stopped displacements and weights ensure that the last stochastic integrals are square-integrable martingales, so their expectations vanish. Since $\mathbf{1}_{\{s<\sigma_R\}} Z_s \leq Z_{s\wedge\sigma_R}$, it follows that

$$\mathbb{E} Z_{t\wedge\sigma_R} \leq C_{R,T,m_0} \int_0^t \mathbb{E} Z_{s\wedge\sigma_R} \, \mathrm{d}s.$$

Gronwall's inequality gives $Z_{t\wedge\sigma_R} = 0$. Since $\sigma_R \to \infty$ almost surely, the representations yield equality of the two measure paths at every fixed time. Equality on rational times and narrow continuity make them indistinguishable. This proves pathwise uniqueness for the weak-solution class under the stated C_b^3 assumption.

4.3. Construction of a strong solution on the prescribed Brownian space

Choose the deterministic positive atomic approximations of Lemma 3.1, and solve their global particle equations on the prescribed Brownian space, all driven by the given W. The mass bounds, weighted compact-containment estimate and estimates for test-function increments proved above give tightness of their continuous measure paths. For any two subsequences, take a joint weak limit of (m^{n_j}, m^{n_ℓ}, W). Apply the martingale-product argument to each measure component, with increment tests depending on the joint past of all components. The cross variations identify both stochastic integrals with the same limiting W, and the Brownian quadratic products show that W is Brownian in this joint filtration. Thus both limiting measures solve the weak equation with initial condition m_0 on one stochastic basis.

By pathwise uniqueness, every such joint limit is supported on the diagonal. The diagonal convergence criterion [24, Lemma 1.1] therefore gives convergence of the whole particle sequence in probability in $C([0,T];\mathcal{M}_+(\mathbb{R}))$ for every finite T. Choose a subsequence that converges almost surely on all integer time intervals simultaneously. Its limit m has narrowly continuous paths and is adapted to the completed prescribed Brownian filtration, since each m_t is a limit in probability of adapted random variables.

The weak equation holds on the prescribed space with its given W. For each test function, the drift and diffusion coefficients are continuous in the narrow topology and are bounded by fixed polynomials in the mass. The uniform mass moments make the drift coefficients and the squares of the diffusion coefficients uniformly integrable. Together with convergence in probability, this gives convergence of the drift in $L^1(\Omega\times[0,T])$ and of each diffusion coefficient in $L^2(\Omega\times[0,T])$. The Itô isometry then gives convergence of the stochastic integrals. Using a countable dense family of test functions and then the test-function bounds yields the full weak formulation. Fatou's lemma gives all required mass moments.

This constructs a global probabilistically strong solution on the prescribed space, with the characteristic representation proved above. On the stochastic basis of any other weak solution, the same particle construction converges to a strong solution driven by its Brownian motion. Pathwise uniqueness identifies the two solutions there. Since the particle laws depend only on the deterministic initial data and the Brownian law, uniqueness in law follows, completing Theorem 1.1. For $m_0 = 0$, the mass equation directly gives the zero solution.

Corollary 4.2 (Continuous dependence under narrow convergence)**.** *Under the assumptions of Theorem 1.1, let $m_0^n \to m_0$ narrowly, and construct the corresponding solutions on a common Brownian space. Then, for each $T<\infty$,*

$$\begin{aligned} m^n &\longrightarrow m && \text{in probability in } C([0,T];\mathcal{M}_+(\mathbb{R})),\\ G*m^n &\longrightarrow G*m && \text{in probability in } C([0,T];H^1). \end{aligned}$$

In particular, the full sequence of positive atomic approximations converges in probability on the prescribed Brownian space. For every bounded narrowly continuous functional F, $m_0\mapsto \mathbb{E}F(m_t)$ is narrowly continuous.

Proof. The initial measures have bounded masses and are uniformly tight. Choose the radii in Lemma 3.1 so that the tail estimate is uniform over this entire initial sequence and its limit. This gives one coercive weight V with bounded initial integrals.

The mass estimate (2.9) holds for every solution of Definition 2.1, with the same dependence on its initial mass. Indeed, testing with 1 gives

$$\mathrm{d}M_t = A_t\,\mathrm{d}t + \sum_k C_{k,t}\,\mathrm{d}W_t^k, \quad |A_t|\le C\varepsilon^2 M_t, \quad \sum_k |C_{k,t}|^2 \le C\varepsilon^2 M_t^2.$$

Here the deterministic CH drift vanishes by antisymmetry, and the remaining coefficients are exactly those in (2.10). After stopping at mass levels, Itô's formula for M^p, followed by BDG, Young's inequality and Gronwall's inequality, gives

$$\mathbb{E}\sup_{t\le T} M_t^p \le C_{p,\xi,\varepsilon,T} M_0^p, \quad p\ge 2.$$

Fatou's lemma removes the stopping. Thus bounded initial masses give the uniform mass moments needed below, independently of whether the momenta are atomic.

The weighted moment estimate also applies to arbitrary solutions of Definition 2.1. Write

$$V_L(x) = 1 + \sum_{j=1}^{L} \chi(x/R_j).$$

These functions are bounded and smooth, increase to V, and have uniform bounds on every positive-order derivative. A spatial cutoff argument, using the mass moments, extends the weak equation from compactly supported tests to V_L. The estimates for its nonlinear drift and its noise are

$$|F_{V_L}(m)| \le CM^2, \quad |B_kV_L| + |B_k^2V_L| \le CV_L,$$

with constants independent of L. Applying Itô's formula up to the stopping times, followed by the BDG inequality and Gronwall's inequality, bounds $\mathbb{E}\sup_{t\le T}\langle V_L, m_t\rangle^p$ uniformly in L. Monotone convergence and Fatou's lemma then give the same bound with V. For the present sequence, the constants are uniform because the initial weighted integrals and masses are bounded.

The compact-containment and test-increment estimates thus hold for these solutions whether or not the data are atomic. Take joint limits of $(m^{n_j}, m^{n_\ell}, m, W)$ and repeat the joint martingale argument above. All limiting measure components have initial condition m_0 and are driven by the same Brownian motion in their joint filtration. Pathwise uniqueness implies that the joint limits are supported on the diagonal. The diagonal criterion gives the claimed convergence in probability.

From any subsequence, the measure-path convergence in probability gives a further subsequence converging almost surely. For each such path, $m \mapsto G * m$ is uniformly continuous on the compact closure of the union of the approximating ranges and the limiting range. Hence the velocities converge in $C([0,T]; H^1)$ along this further subsequence. The subsequence criterion yields convergence in probability for the full sequence. Finally, boundedness of F gives convergence of its expectations. □

5. A cubic functional and proof of H^1 stability

To obtain stability in terms of the initial velocity error, the estimates below must be independent of the perturbation's total momentum. Positivity allows proximity to the peakon family to be measured by a defect from rank one. With T_m denoting the positive Green operator defined below, the argument combines the following relations:

$$d_{\mathcal{P}}(u)^4 \le 4\sqrt{2}\,\mathcal{Z}(u), \quad 6\mathcal{J}(u) = \Phi(T_m),$$
$$T_m^{(j)} = T_m^{1/2} K_j T_m^{1/2}, \quad |\mathcal{Z}_k^{(j)}| \lesssim \mathcal{Z}, \quad j = 1, 2.$$

Here $T_m^{(j)}$ denotes variation along one deterministic noise flow, and Lemma 5.5 supplies the bounded operators K_j. Static coercivity converts control of the defect into an H^1 distance estimate. The relative operator bounds control the drift and diffusion of the defect in terms of the defect itself, with constants independent of total momentum. To apply these bounds to general measure solutions, the cubic balance is first established on each solution's original probability space.

5.1. Static coercivity at the peakon family

Proposition 5.1 (Distance estimate independent of total momentum)**.** *Let $u = G * m$ be nonzero, where $m \ge 0$ is finite. Then*

$$\mathcal{J}(u) \ge 0, \quad \operatorname{dist}_{H^1}(u, \mathcal{P})^4 \le 4\sqrt{2}\,\sqrt{E(u)}\,\mathcal{J}(u). \tag{5.1}$$

For each fixed $c > 0$ there are $\delta_c, C_c > 0$ such that

$$\|u - ce^{-|\cdot - q|}\|_{H^1} \le \delta_c \quad \Longrightarrow \quad |\mathcal{J}(u)| \le C_c \|u - ce^{-|\cdot - q|}\|_{H^1}^2.$$

The constants are independent of the total momentum and of q.

Proof. The velocity u is positive and continuous and vanishes at infinity. Hence $S = \max u > 0$, and the maximum is attained at some q. By the reproducing identity,

$$\|u - ae^{-|\cdot - y|}\|_{H^1}^2 = E + 2a^2 - 4au(y).$$

Minimization first over $a > 0$ and then over y yields $d_{\mathcal{P}}(u)^2 = E - 2S^2$. Set $g = u_x - u$ on $x < q$ and $g = u_x + u$ on $x > q$. Integration by parts gives

$$\int_{\mathbb{R}} g^2\,\mathrm{d}x = E - 2S^2, \quad \int_{\mathbb{R}} ug^2\,\mathrm{d}x = \mathcal{F} - \tfrac{4}{3}S^3.$$

The bound $u \le S$ gives $\mathcal{F} \le SE - \frac{2}{3}S^3$. This is the standard peakon inequality used in [17]. Put $z = S/\sqrt{E/2} \in [0,1]$. Then

$$\mathcal{J} \ge \frac{E^{3/2}}{3\sqrt{2}}(1-z)^2(z+2), \quad \operatorname{dist}_{H^1}(u, \mathcal{P})^2 = E(1 - z^2).$$

For $0 \le z \le 1$ one has

$$\frac{4}{3}(z+2) - (1+z)^2 = \frac{(1-z)(3z+5)}{3} \ge 0.$$

Multiplying the lower bound for $\mathcal{J}$ by $4\sqrt{2}\sqrt{E}$ gives (5.1), also when $z = 1$. The functional $\mathcal{F}$ is a continuous cubic polynomial on H^1, and $\mathcal{J}$ is twice continuously differentiable near every nonzero peakon. At $\varphi = ce^{-|\cdot - q|}$,

$$E'(\varphi)[w] = 4c\, w(q), \quad \mathcal{F}'(\varphi)[w] = 4c^2 w(q), \quad \mathcal{J}(\varphi) = \mathcal{J}'(\varphi) = 0.$$

For the first variation of $\mathcal{F}$, the distributional identity

$$3\varphi^2 + \varphi_x^2 - 2(\varphi\varphi_x)_x = 4c^2\delta_q$$

justifies integration by parts across the peak. The second derivative of $\mathcal{J}$ is bounded on an H^1 ball whose radius depends only on c. Taylor's integral formula gives the quadratic bound. Translation invariance makes the constants independent of q, and the argument uses no bound on the total momentum. □

Remark 5.2 (Sharpness of the distance estimate). Let $p > 0$, $r > 0$, $m_r = p\delta_0 + p\delta_r$ and $u_r = G * m_r$. Writing $s = e^{-r}$, direct calculation gives

$$E(u_r) = p^2(1+s), \quad \mathcal{F}(u_r) = \frac{p^3}{3}(1+3s), \quad \max u_r = \frac{p}{2}(1+s).$$

Consequently,

$$d_{\mathcal{P}}(u_r)^2 = \frac{p^2}{2}(1-s^2), \quad \mathcal{Z}(u_r) = \frac{p^4}{3}\left[\sqrt{2}(1+s)^2 - (1+3s)\sqrt{1+s}\right].$$

As $r \downarrow 0$, these expressions yield

$$d_{\mathcal{P}}(u_r)^2 = p^2 r + O(r^2), \quad \mathcal{Z}(u_r) = \frac{p^4}{4\sqrt{2}}r^2 + O(r^3), \quad \frac{d_{\mathcal{P}}(u_r)^4}{\mathcal{Z}(u_r)} \longrightarrow 4\sqrt{2}.$$

Thus the constant $4\sqrt{2}$ in (5.1) is sharp. Moreover, no estimate $d_{\mathcal{P}}(u)^\alpha \le C\mathcal{Z}(u)$ with $0 < \alpha < 4$ and a fixed finite C can hold in a neighborhood of the limiting peakon $pe^{-|\cdot|}$. The fourth power in this coercivity estimate is therefore optimal.

5.2. A three-point kernel and its weak stochastic balance

Set

$$A(m) = \iint_{\mathbb{R}^2} e^{-|x-y|} m(\mathrm{d}x) m(\mathrm{d}y) = 2E(u_m), \quad H_n(\boldsymbol{x}) = e^{-(\max_i x_i - \min_i x_i)} \quad (n = 2, 3).$$

The cubic functional has the representation

$$B(m) := \int_{\mathbb{R}^3} H_3 \, \mathrm{d}m^{\otimes 3} = 6\mathcal{F}(u_m), \quad 6\mathcal{J}(u_m) = A(m)^{3/2} - B(m). \tag{5.2}$$

Indeed, integration by parts gives $\mathcal{F} = \frac{1}{2}\int_{\mathbb{R}} u^2 \,\mathrm{d}m + \frac{1}{2}\int_{\mathbb{R}} u^3 \,\mathrm{d}x$. For kernel arguments ordered as $x \le y \le z$, set $a = y - x$, $b = z - y$. The symmetrized kernel of the first term is

$$\frac{1}{24}(e^{-2a-b} + e^{-a-b} + e^{-a-2b}),$$

whereas direct integration over the four intervals determined by x, y, z gives the kernel of the second term

$$\frac{1}{8}e^{-a-b} - \frac{1}{24}(e^{-2a-b} + e^{-a-2b}).$$

The sum equals $H_3/6$. We extend this identity to measures with atoms by approximation with smooth finite measures, using strong H^1 convergence of the corresponding velocities and continuity of the cubic polynomial.

For a single noise coefficient, define the tensor operator

$$\mathcal{T}_\xi^{(n)} = \sum_{i=1}^n \xi(x_i)\partial_i - \sum_{i=1}^n \xi'(x_i).$$

With $X = \max_i x_i, Y = \min_i x_i$, write

$$a_\xi = \xi(X) - \xi(Y), \quad b_\xi = \sum_i \xi'(x_i), \quad w_\xi = \xi(X)\xi'(X) - \xi(Y)\xi'(Y), \quad d_\xi = \sum_i \xi(x_i)\xi''(x_i).$$

Differentiation in each ordering region gives

$$\mathcal{T}_\xi^{(n)} H_n = -(a_\xi + b_\xi)H_n, \quad (\mathcal{T}_\xi^{(n)})^2 H_n = \big((a_\xi + b_\xi)^2 - w_\xi - d_\xi\big)H_n. \tag{5.3}$$

The right-hand sides are bounded continuous functions on all of $\mathbb{R}^n$, including the collision diagonals. Their values need not vanish on the full diagonal; at a single peakon, the cancellation occurs in the combination defining $\mathcal{J}$. Use these extensions to define

$$A_k' = \int_{\mathbb{R}^2} \mathcal{T}_{\xi_k}^{(2)} H_2 \, \mathrm{d}m^{\otimes 2}, \quad A_k'' = \int_{\mathbb{R}^2} (\mathcal{T}_{\xi_k}^{(2)})^2 H_2 \, \mathrm{d}m^{\otimes 2},$$

and define B_k', B_k'' analogously with $H_3, m^{\otimes 3}$. Primes denote derivatives in the deterministic noise direction.

Lemma 5.3 (Cubic balance for every weak solution)**.** *On the original probability space and with the original Brownian motions, every solution in the stated class satisfies*

$$\mathrm{d}A = \varepsilon \sum_k A_k' \, \mathrm{d}W^k + \frac{\varepsilon^2}{2} \sum_k A_k'' \, \mathrm{d}t,$$
$$\mathrm{d}B = \varepsilon \sum_k B_k' \, \mathrm{d}W^k + \frac{\varepsilon^2}{2} \sum_k B_k'' \, \mathrm{d}t.$$

Proof. The first identity follows from (3.12), since $A = 2E$. For the cubic identity, let us first derive the tensor formula for smooth kernels and then justify its use at the collision diagonals of H_3. For a bounded smooth three-point test h with bounded derivatives through order two, this formula reads

$$\mathrm{d}\int_{\mathbb{R}^3} h \, \mathrm{d}m^{\otimes 3} = \int_{\mathbb{R}^3} \mathcal{L}_m h \, \mathrm{d}m^{\otimes 3} \, \mathrm{d}t + \varepsilon \sum_k \int_{\mathbb{R}^3} \mathcal{T}_{\xi_k}^{(3)} h \, \mathrm{d}m^{\otimes 3} \, \mathrm{d}W^k$$
$$+ \frac{\varepsilon^2}{2} \sum_k \int_{\mathbb{R}^3} (\mathcal{T}_{\xi_k}^{(3)})^2 h \, \mathrm{d}m^{\otimes 3} \, \mathrm{d}t, \quad \mathcal{L}_m = \sum_i u(x_i)\partial_i - \sum_i \bar{u}_x(x_i).$$

First apply the scalar product rule to finite sums of products of one-point tests, then pass to general kernels by compact C^2 approximation and spatial cutoffs. The nonlinear integrand is bounded by CM^4, and the noise coefficients are bounded by CM^3. The mass moment bounds therefore justify passage to the limit. The specified representative $\bar{u}_x = G' * m$, $G'(0) = 0$, is retained.

The deterministic contribution to the H_3 balance vanishes, even for atomic momentum. Indeed, its directional derivative is

$$\mathcal{L}_m H_3 = -H_3 \left(u(X) - u(Y) + \sum_i \bar{u}_x(x_i) \right).$$

After symmetrization over four integration variables, the required identity is

$$\sum_{\ell=1}^4 H_{-\ell} \left[G(X_{-\ell} - r_\ell) - G(Y_{-\ell} - r_\ell) + \sum_{i \neq \ell} G'(r_i - r_\ell) \right] = 0. \tag{5.4}$$

Here $H_{-\ell}, X_{-\ell}, Y_{-\ell}$ are defined using the three points that remain after r_ℓ is omitted. For $r_1 < r_2 < r_3 < r_4$ and successive gaps a, b, c, put $R = e^{-a-b-c}$ and $S = e^{-a-2b-c}$. The four summands are $-R - S/2, -S/2, S/2, R + S/2$, respectively. For coinciding points, pair the derivative terms as

$$\sum_{\ell<i} G'(r_i - r_\ell)(H_{-\ell} - H_{-i}).$$

Their coefficients vanish at the corresponding collisions. The symmetrized expression is continuous and agrees at collisions with the value obtained using $G'(0) = 0$. Hence (5.4) holds everywhere, and its integral vanishes.

It remains to justify the tensor formula for the nonsmooth kernel. Let $H_\rho = \rho^{-3}\chi(\cdot/\rho) * H_3$ for a compactly supported smooth mollifier χ on $\mathbb{R}^3$. Then, uniformly in the three variables,

$$H_\rho \to H_3, \quad \mathcal{T}^{(3)}_\xi H_\rho \to \mathcal{T}^{(3)}_\xi H_3, \quad (\mathcal{T}^{(3)}_\xi)^2 H_\rho \to (\mathcal{T}^{(3)}_\xi)^2 H_3.$$

To prove the last convergence, write $H_3 = \exp(-\frac{1}{2}\sum_{i<j}|x_i - x_j|)$. Its distributional Hessian has a bounded piecewise smooth part; its singular part on $x_i = x_j$ is a bounded coefficient times the squared normal $(e_i - e_j)^{\otimes 2}$ and hyperplane measure. No additional second-derivative measure occurs on triple intersections. Contracting with the noise vector gives a factor $(\xi(x_i) - \xi(x_j))^2$. Wherever the mollifier intersects that hyperplane, this factor is $O(\rho^2)$, whereas convolution of the hyperplane measure is $O(\rho^{-1})$. The singular contribution is therefore uniformly $O(\rho)$. For the regular part, replacing the coefficients at x by their values at the convolution variable gives an $O(\rho)$ error. The resulting kernel is the continuous expression in (5.3), which is uniformly continuous under the C^3_b assumption. The same commutator estimate holds for the first-order and zeroth-order terms.

For the deterministic direction, $\|u\|_\infty$ and $\mathrm{Lip}(u)$ are at most $M/2$. Commuting $u(x_i)$ through the convolution and using the continuous directional formula for H_3 gives a uniform $C\rho M$ error. Multiplying $H_\rho - H_3$ by the bounded, possibly discontinuous $\bar{u}_x$ gives the same bound. Consequently

$$\left|\int_{\mathbb{R}^3} \mathcal{L}_m H_\rho \, \mathrm{d}m^{\otimes 3}\right| \le C\rho M^4.$$

The mass moment bounds, the dominated convergence theorem and Itô isometry justify passage to the limit on the given probability space. □

For nonzero initial momentum, $A_t > 0$ at every finite time, since the mass remains positive and the kernel defining A is strictly positive. Continuity permits localization by both A and A^{-1}. Applying the scalar Itô formula to (5.2) therefore gives

$$\mathrm{d}\mathcal{J} = \varepsilon \sum_k \mathcal{J}'_k \, \mathrm{d}W^k + \frac{\varepsilon^2}{2}\sum_k \mathcal{J}''_k \, \mathrm{d}t, \tag{5.5}$$

where

$$\mathcal{J}'_k = \frac{1}{6}\left(\frac{3}{2}\sqrt{A}\,A'_k - B'_k\right), \quad \mathcal{J}''_k = \frac{1}{6}\left(\frac{3}{2}\sqrt{A}\,A''_k + \frac{3(A'_k)^2}{4\sqrt{A}} - B''_k\right).$$

These coefficients are the first and second ordinary derivatives along the deterministic coadjoint flow of ξ_k. Since the calculation uses scalar momentum functionals, it requires no differentiability of translated peakons as H^1-valued curves.

5.3. A trace functional and its derivatives

Let $\mathcal{H}$ be $H^1(\mathbb{R})$ with squared norm

$$\|f\|^2_{\mathcal{H}} = \int_{\mathbb{R}} (f_x^2 + f^2/4)\,\mathrm{d}x.$$

Its reproducing kernel is $k(x, y) = e^{-|x-y|/2}$, the Green kernel of $-\partial_x^2 + 1/4$. The evaluation map $j : \mathcal{H} \to L^2(m)$ is Hilbert–Schmidt: for any orthonormal basis, Tonelli's theorem and the reproducing identity give $\|j\|^2_{\mathrm{HS}} = \int_{\mathbb{R}} k(x, x)\,\mathrm{d}m = M$. Thus

$$(T_m f)(x) = \int_{\mathbb{R}} e^{-|x-y|/2} f(y) m(\mathrm{d}y) = jj^* f$$

is a positive trace-class operator. Kernel multiplication yields

$$\operatorname{tr}(T_m^2) = A, \quad \operatorname{tr}(T_m^3) = B, \quad 6\mathcal{J} = \Phi(T_m), \quad \Phi(T) = (\operatorname{tr} T^2)^{3/2} - \operatorname{tr} T^3. \tag{5.6}$$

For the cubic identity, we use $|x-y| + |y-z| + |z-x| = 2(\max - \min)$. Nonnegative momentum makes T_m positive. For a nonzero positive operator, Φ vanishes precisely at rank one. The next lemma measures the remaining spectrum in Hilbert–Schmidt norm and bounds the derivatives of Φ by Φ itself. The constants do not depend on the trace, which equals the total momentum in the present setting.

Lemma 5.4 (Relative derivative bounds for the rank-one defect)**.** *Let $T \neq 0$ be a positive trace-class operator and $H = T^{1/2} K T^{1/2}$, where K is self-adjoint and $\|K\| \le \kappa$. Write $\lambda_1 \ge \lambda_2 \ge \cdots \ge 0$ for the eigenvalues, $r = (\sum_i \lambda_i^2)^{1/2}$ and $S = \sum_{i\ge 2} \lambda_i^2$. Then*

$$\tfrac{1}{2} rS \le \Phi(T) \le \tfrac{3}{2} rS. \tag{5.7}$$

In particular, $\Phi(T) = 0$ if and only if T has rank one. Moreover,

$$|D\Phi(T)[H]| \le 3\kappa\Phi(T), \quad |D^2\Phi(T)[H,H]| \le 72\kappa^2\Phi(T).$$

Proof. Set $a = \lambda_1$, so that $S = r^2 - a^2$. We first prove (5.7). If $S = 0$, both sides are zero. For $S > 0$, the lower bound follows from $\sum_i \lambda_i^3 \le a r^2$ and $r^2(r-a) = r^2 S/(r+a) \ge rS/2$. The upper bound follows by discarding the nonnegative cubes of the remaining eigenvalues and using

$$\frac{r^3 - a^3}{r(r^2 - a^2)} = \frac{r^2 + ra + a^2}{r(r+a)} \le \frac{3}{2},$$

where the last inequality is equivalent to $(r-a)(r+2a) \ge 0$. In an eigenbasis,

$$D\Phi(T)[H] = 3 \sum_i \lambda_i (r - \lambda_i) H_{ii}.$$

Since $|H_{ii}| \le \kappa\lambda_i$ and all coefficients $\lambda_i(r-\lambda_i)$ are nonnegative, this proves the first inequality.

The Hilbert–Schmidt Hessian is

$$D^2\Phi(T)[H,H] = 3r \operatorname{tr} H^2 + \frac{3}{r}(\operatorname{tr} TH)^2 - 6\operatorname{tr}(TH^2).$$

Decompose $H = \left(\begin{smallmatrix} h & z^* \\ z & D \end{smallmatrix}\right)$ relative to a top eigenvector and its orthogonal complement, and write $C = \operatorname{diag}(\lambda_2, \dots)$ and $b = \operatorname{tr}(CD)$. One third of the Hessian equals

$$\frac{(r-a)^2}{r} h^2 + \frac{2a}{r} hb + \frac{b^2}{r} + 2 \sum_{j\ge 2} (r - a - \lambda_j)|z_j|^2 + r \operatorname{tr} D^2 - 2\operatorname{tr}(CD^2).$$

Here $|h| \le \kappa a$, $|b| \le \kappa S$ and $\operatorname{tr} D^2 \le \kappa^2 S$. The latter follows from $D_{ij} = \sqrt{\lambda_i \lambda_j} K_{ij}$ and $\lambda_i\lambda_j \le (\lambda_i^2 + \lambda_j^2)/2$. The absolute values of the first three terms are bounded, respectively, by $\kappa^2 rS, 2\kappa^2 rS, \kappa^2 rS$. For the off-diagonal term, $|z_j|^2 = a\lambda_j |K_{1j}|^2$ and $\sum_j |K_{1j}|^2 \le \kappa^2$ give

$$\begin{aligned} 2 \sum_{j\ge 2} |r - a - \lambda_j|\, |z_j|^2 &\le 2a(r-a) \sum_{j\ge 2} \lambda_j |K_{1j}|^2 + 2a \sum_{j\ge 2} \lambda_j^2 |K_{1j}|^2 \\ &\le 2\kappa^2 rS + 2\kappa^2 aS \le 4\kappa^2 rS. \end{aligned}$$

The last estimate uses $a(r-a) \le S$, $\lambda_j \le r$ and $\lambda_j^2 \le S$. Since $0 \le C \le rI$, the last two terms satisfy

$$|r \operatorname{tr}(D^2) - 2\operatorname{tr}(CD^2)| \le 3r \operatorname{tr}(D^2) \le 3\kappa^2 rS.$$

The absolute value of the Hessian is thus at most $33\kappa^2 rS$, which is at most $66\kappa^2\Phi(T)$ by (5.7). This proves the stated bound with 72. The argument also covers repeated and zero eigenvalues. If $S = 0$, all the displayed terms vanish. In infinite dimension, let P_j project onto the first j eigenvectors of T, and apply the finite-dimensional argument to $T_j = P_j T P_j$ and $H_j = P_j H P_j$. The compressed operator $P_j K P_j$ has norm at most κ. Also $T_j \to T$ and $H_j \to H$ in Hilbert–Schmidt norm. Products of two sequences converging in Hilbert–Schmidt norm converge in trace norm, and convergence in Hilbert–Schmidt norm also implies convergence in operator norm. Thus all traces in the derivative formulas converge, and $\|T_j\|_{\mathrm{HS}} \to r > 0$. Passing to the limit gives the stated inequalities without a spectral-gap assumption. □

5.4. Noise deformation of the Green operator

Fix one $\xi \in C_b^3$ and consider its deterministic flow $\partial_s \phi_s = \xi(\phi_s)$, $\phi_0 = \mathrm{id}$, with Jacobian $J_s = \partial_x \phi_s$. The coadjoint action on momentum is $m_s = (\phi_s)_\#(J_s^{-1} m)$. The operator T_{m_s} is unitarily equivalent to the following operator on the fixed space $L^2(m)$:

$$T_s(x,y) = \frac{\exp(-|\phi_s(x) - \phi_s(y)|/2)}{\sqrt{J_s(x) J_s(y)}}. \tag{5.8}$$

The unitary map sends $g \in L^2(m_s)$ to $g(\phi_s(\cdot))/\sqrt{J_s(\cdot)}$. Order preservation keeps the sign of each spatial difference fixed. The kernels in (5.8), including their diagonal values, are therefore twice continuously differentiable in s, with locally uniform bounds on their derivatives. Finite mass and dominated convergence give C^2 regularity in Hilbert–Schmidt norm on the fixed space $L^2(m)$.

Lemma 5.5 (Relative bounds for derivatives of the Green operator)**.** *At $s = 0$, there are bounded self-adjoint operators K_1, K_2 on $L^2(m)$ such that*

$$T_0' = T_m^{1/2} K_1 T_m^{1/2}, \quad T_0'' = T_m^{1/2} K_2 T_m^{1/2}, \quad \|K_1\| \le \kappa_1, \quad \|K_2\| \le \kappa_2,$$

with κ_1, κ_2 given by (1.9) for ξ.

Proof. For real $f \in L^2(m)$, the signed measure $f\, m$ has finite total variation. Put

$$n_s = (\phi_s)_\#(J_s^{-1/2} f\, m), \quad v_s = (-\partial_x^2 + 1/4)^{-1} n_s, \quad Q_s = \langle f, T_s f \rangle = \|v_s\|_{\mathcal{H}}^2.$$

The signed measure evolves in the distributional sense according to $\partial_s n_s = -\xi \partial_x n_s - \frac{3}{2}\xi' n_s$. For smooth densities, integration by parts gives the cancellation

$$Q_s' = 2 \int_{\mathbb{R}} (\xi v_{s,x} - \tfrac{1}{2}\xi' v_s)(v_s/4 - v_{s,xx})\, \mathrm{d}x = \frac{1}{2} \int_{\mathbb{R}} (\xi''' - \xi') v_s^2\, \mathrm{d}x. \tag{5.9}$$

The calculation is first justified for compactly supported smooth densities and then extended to finite signed measures by truncation and convolution. The approximating measures have uniformly bounded total variations, are uniformly tight, and their potentials converge in $\mathcal{H}$. The differentiated kernel on the left is bounded and continuous, since its apparent diagonal jump is multiplied by $\xi(x) - \xi(y)$. Thus (5.9) also holds for the present signed measures. In particular, $|Q_s'| \le 2(L_1 + L_3) Q_s$.

The curve v_s is differentiable in $L^2(\mathbb{R})$: it is given by the Bochner integral of the translated Green kernel against a finite signed measure, and translations of this kernel are C^1 in L^2. The time-dependent weights and shifts have locally bounded derivatives, so differentiation under the integral is justified. For $g \in L^2$ with $\|g\|_2 \le 1$, let $w = (-\partial_x^2 + 1/4)^{-1} g$. Fourier multipliers give $\|w\|_2 \le 4$, $\|w_x\|_2 \le 1$, $\|w_{xx}\|_2 \le 1$. Distributional duality gives

$$\langle \partial_s v_s, g \rangle = \langle v_s, \xi w_x - \tfrac{1}{2}\xi' w \rangle_{\mathcal{H}}.$$

Put $h = \xi w_x - \frac{1}{2}\xi' w$. Since $h_x = \xi w_{xx} + \frac{1}{2}\xi' w_x - \frac{1}{2}\xi'' w$, the multiplier bounds yield

$$\|h_x\|_2 \le L_0 + \tfrac{1}{2} L_1 + 2L_2, \quad \tfrac{1}{2}\|h\|_2 \le \tfrac{1}{2} L_0 + L_1.$$

Consequently

$$\|h\|_{\mathcal{H}} \le \|h_x\|_2 + \tfrac{1}{2}\|h\|_2 \le \tfrac{3}{2} L_0 + \tfrac{3}{2} L_1 + 2L_2 \le 2(L_0 + L_1 + L_2).$$

Hence $\|\partial_s v_s\|_2 \le 2(L_0 + L_1 + L_2)\|v_s\|_{\mathcal{H}}$. Differentiating the right-hand side of (5.9) in L^2 now yields

$$Q_s'' = \int_{\mathbb{R}} (\xi''' - \xi') v_s\, \partial_s v_s\, \mathrm{d}x, \quad |Q_s''| \le 4(L_1 + L_3)(L_0 + L_1 + L_2) Q_s.$$

These bounds give the quadratic-form inequalities $-\kappa_i T_m \le T_0^{(i)} \le \kappa_i T_m$. For the factorization, define a symmetric form on $\operatorname{ran} T_m^{1/2}$ by $\mathfrak{b}(T_m^{1/2} f, T_m^{1/2} g) = \langle f, T_0^{(i)} g\rangle$. The diagonal estimate and polarization imply

$$|\langle f, T_0^{(i)} g\rangle| \le \kappa_i \|T_m^{1/2} f\| \, \|T_m^{1/2} g\|.$$

Indeed, apply the polarization identity to tf and $t^{-1}g$ and optimize over $t > 0$. In particular, this pairing vanishes if either vector lies in $\ker T_m^{1/2}$, so the form is well defined. It extends to the closure of the range, where it is represented by a bounded self-adjoint operator of norm at most κ_i. Extending that operator by zero on the orthogonal complement proves the asserted factorization. □

The Green-operator factorizations place each noise variation in the relative form required by Lemma 5.4. Together with (5.6), they give the following bounds for every nonzero finite positive momentum:

$$\begin{aligned} &|\mathcal{J}_k'| \le 3\kappa_{1,k}\mathcal{J}, \quad |\mathcal{J}_k''| \le (72\kappa_{1,k}^2 + 3\kappa_{2,k})\mathcal{J}, \\ &|E_k'| \le 2\kappa_{1,k}E, \quad |E_k''| \le 2(\kappa_{1,k}^2 + \kappa_{2,k})E. \end{aligned} \tag{5.10}$$

For the energy bounds, we use

$$E' = \operatorname{tr}(TT'), \quad E'' = \operatorname{tr}((T')^2) + \operatorname{tr}(TT'').$$

The first factorization gives $\|T'\|_{\mathrm{HS}} \le \kappa_1 \|T\|_{\mathrm{HS}}$. The second factorization and cyclicity of the trace give $|\operatorname{tr}(TT'')| = |\operatorname{tr}(K_2 T^2)| \le \kappa_2 \operatorname{tr}(T^2)$. Since $\operatorname{tr}(T^2) = 2E$, these bounds prove (5.10). All constants are independent of M.

5.5. Proof of Theorem 1.3

Common defect balance. The scalar product formula applied to (5.5) and the energy balance gives

$$\mathrm{d}\mathcal{Z} = \varepsilon \sum_k \mathcal{Z}_k' \, \mathrm{d}W^k + \frac{\varepsilon^2}{2} \sum_k \mathcal{Z}_k'' \, \mathrm{d}t, \quad |\mathcal{Z}_k'| \le 4\kappa_{1,k}\mathcal{Z}, \quad |\mathcal{Z}_k''| \le (80\kappa_{1,k}^2 + 4\kappa_{2,k})\mathcal{Z}. \tag{5.11}$$

Indeed, for $r = \sqrt{E}$ the bounds are $|r_k'| \le \kappa_{1,k} r$ and $|r_k''| \le (2\kappa_{1,k}^2 + \kappa_{2,k})r$. The second derivative of $r\mathcal{J}$ consists of the two unmixed terms and the cross term $2r_k'\mathcal{J}_k'$. These bounds give the constants in (5.11). For each solution, its mass moment bounds justify localization. The drift bound in (5.11) makes $e^{-\lambda_Z t}\mathcal{Z}_t$ a nonnegative local supermartingale, and hence a supermartingale. Moreover, $0 \le \mathcal{Z} \le (\sqrt{2}/3)E^2 \le (\sqrt{2}/12)M^4$ gives finite-time integrability directly. These mass moments justify the stochastic calculation; the relative coefficient bounds used in the resulting estimate are independent of the mass.

(i) Initial H^1 perturbations. By Proposition 5.1, exit from the peakon neighborhood requires $\mathcal{Z}_t \ge b_\eta$. The velocity has continuous H^1 paths by Lemma 2.3, and distance to any subset of a metric space is a Lipschitz function. Thus the distance process is continuous and the hitting times used here are stopping times. On $\{\tau_\eta \le T\}$,

$$\sup_{t \le T} e^{-\lambda_Z t}\mathcal{Z}_t \ge e^{-\lambda_Z \tau_\eta} b_\eta \ge e^{-\lambda_Z T} b_\eta.$$

The maximal inequality for the nonnegative supermartingale proves (1.6). It remains to estimate $\mathcal{Z}(u_0)$ from the initial velocity error. Set $\varphi = ce^{-|\cdot - q_0|}$ and $w = u_0 - \varphi$. The vanishing first variation of $\mathcal{J}$ at φ reduces this estimate to a bound on its second derivative. On H^1, the sharp embedding $\|f\|_\infty \le \|f\|_{H^1}/\sqrt{2}$ gives

$$|D^2\mathcal{F}(u)[w,w]| = \left| \int_{\mathbb{R}} (6uw^2 + 2uw_x^2 + 4ww_x w_x)\,\mathrm{d}x \right| \le 6\sqrt{2}\,\|u\|_{H^1}\|w\|_{H^1}^2.$$

The second derivative of $(\sqrt{2}/3)\|u\|_{H^1}^3$ has operator norm at most $2\sqrt{2}\|u\|_{H^1}$, with the same bound for the continuous extension at $u = 0$. At the reference peakon, $\mathcal{J}(\varphi) = D\mathcal{J}(\varphi) = 0$. Each point on the segment $\varphi + sw$, $0 \le s \le 1$, has H^1 norm at most $\sqrt{2}c + \delta$. The two Hessian bounds and Taylor's integral formula give

$$\begin{aligned} 0 \le \mathcal{J}(u_0) &= \int_0^1 (1-s) D^2\mathcal{J}(\varphi + sw)[w,w]\,\mathrm{d}s \\ &\le 4\sqrt{2}(\sqrt{2}c + \delta)\delta^2. \end{aligned}$$

If the segment passes through zero, the C^2 extension of $u \mapsto \|u\|_{H^1}^3$ applies there. Multiplying by $\sqrt{E(u_0)} \le \sqrt{2}c + \delta$ gives

$$\mathcal{Z}(u_0) \le 4\sqrt{2}(\sqrt{2}c + \delta)^2\delta^2.$$

Substitution in (1.6) proves (1.5) and completes the proof of assertion (i).

(ii) Defect balance and refined exit probabilities. The supermartingale property and (1.6) have been established. For the refined tail estimate, define progressively measurable coefficients by

$$\alpha_t = \frac{\varepsilon^2}{2\mathcal{Z}_t}\sum_k \mathcal{Z}_k''(m_t), \quad \beta_{t,k} = \frac{\varepsilon\mathcal{Z}_k'(m_t)}{\mathcal{Z}_t} \quad \text{on } \{\mathcal{Z}_t > 0\},$$

and set them to zero on $\{\mathcal{Z}_t = 0\}$. The bounds in (5.11) imply $|\alpha_t| \le \lambda_Z$ and $\sum_k \beta_{t,k}^2 \le \varepsilon^2 V_Z$. They also show that (5.11) is the linear equation $\mathrm{d}\mathcal{Z} = \alpha\mathcal{Z}\,\mathrm{d}t + \mathcal{Z}\sum_k \beta_k\,\mathrm{d}W^k$, including at zero. Uniqueness for this scalar linear equation gives

$$\mathcal{Z}_t = \mathcal{Z}_0 \exp\left\{\int_0^t \left(\alpha_s - \frac{1}{2}\sum_k \beta_{s,k}^2\right)\mathrm{d}s + \sum_k \int_0^t \beta_{s,k}\,\mathrm{d}W_s^k\right\}.$$

When $\mathcal{Z}$ is initially positive, it cannot reach zero in finite time. If $\mathcal{Z}_0 > 0$, it follows that

$$\log(\mathcal{Z}_t/\mathcal{Z}_0) \le \lambda_Z t + N_t, \quad \langle N\rangle_t \le \varepsilon^2 V_Z t, \quad N_t = \varepsilon\sum_k \int_0^t \frac{\mathcal{Z}_k'}{\mathcal{Z}}\,\mathrm{d}W^k.$$

Here $\langle N\rangle_t$ denotes the quadratic variation. Write $v = \varepsilon^2 V_Z > 0$. For $r > 0$ and $\theta > 0$, applying the exponential martingale inequality at the first time N reaches r gives

$$\mathbb{P}\{\sup_{t\le T} N_t \ge r\} \le \exp(-\theta r + \theta^2 vT/2).$$

Choosing $\theta = r/(vT)$ yields $\exp(-r^2/(2vT))$. If $\log(b_\eta/\mathcal{Z}_0) - \lambda_Z T > 0$, an exit by T requires N to reach this value. If this expression is nonpositive, the bound by one suffices. Thus (1.7) holds in both cases.

(iii) Mean exit times and invariance. Let $\sigma = \inf\{t : \mathcal{Z}_t \ge b_\eta\}$. Then $\sigma \le \tau_\eta$. If $\mathbb{E}\sigma = \infty$, (1.8) is immediate. Otherwise $\sigma < \infty$ almost surely. Under the hypotheses of the mean exit-time bound, $\mathcal{Z}_0 < b_\eta$, so continuity gives $\mathcal{Z}_\sigma = b_\eta$. Put $Y_t = e^{-\lambda_Z t}\mathcal{Z}_t$. Optional stopping at $\sigma \wedge t$ and Fatou's lemma give

$$b_\eta \mathbb{E}e^{-\lambda_Z\sigma} = \mathbb{E}\liminf_{t\to\infty} Y_{\sigma\wedge t} \le \liminf_{t\to\infty}\mathbb{E}Y_{\sigma\wedge t} \le \mathcal{Z}_0.$$

Jensen's inequality implies $e^{-\lambda_Z\mathbb{E}\sigma} \le \mathcal{Z}_0/b_\eta$. Taking logarithms and using $\sigma \le \tau_\eta$ gives (1.8). When $\lambda_Z = 0$, either $\varepsilon = 0$ or every coefficient is constant, and $\mathcal{Z}$ is conserved; division by λ_Z is not used in this case.

If $\mathcal{Z}_0 = 0$, the supermartingale property and nonnegativity imply $\mathcal{Z}_t = 0$ almost surely at every rational time. By continuity, this equality holds simultaneously for all times. The optimal amplitude and location in Proposition 5.1 then give a peakon at zero distance from the solution. Strict positivity of the total momentum excludes the zero profile, so the solution remains in $\mathcal{P}$.

This proves assertion (iii) and completes the proof.

Remark 5.6. The initial momentum measures admitted by (1.5) may have arbitrarily large finite total momentum. Only the initial H^1 error, the reference height and the noise coefficients enter the probability bound. The fourth-power distance estimate and the quadratic bound in the initial error together yield the ratio δ^2/η^4. Comparison with a specified multipeakon solution is developed in Part II [41].

6. Exact peakon dynamics and particle illustrations

The exact examples below connect Theorem 1.3 with the evolution of peakon height, position and separation. A single peakon preserves its shape under spatially varying noise, while a small positive second peak separates under spatially constant noise. These examples identify the comparison family and provide exact benchmarks for the computations. The numerical panels then illustrate the distinction between height variation and distance to the peakon family, the initial control of the peakon defect $\mathcal{Z}$, and the response of the computed observables to time-step refinement.

6.1. Amplitude dynamics of a single peakon

If $m_0 = 2a_0\delta_{q_0}$ with $a_0 > 0$, then $D_0 = 0$, so Theorem 2.2 and Lemma 2.3 give $m_t = 2a_t\delta_{q_t}$. Positivity of the mass and narrow continuity ensure that a_t and q_t are continuous. Applying localized coordinate tests in (2.3) then yields

$$\mathrm{d}q_t = a_t\,\mathrm{d}t + \varepsilon\sum_k \xi_k(q_t)\circ\mathrm{d}W_t^k, \quad \mathrm{d}a_t = -\varepsilon a_t\sum_k \xi_k'(q_t)\circ\mathrm{d}W_t^k. \tag{6.1}$$

The one-particle construction gives the same equations.

Take $\xi_1(x) = \sin x$ and $\xi_2(x) = \cos x$. The Itô equation for $\log a_t$ is

$$\mathrm{d}\log a_t = -\varepsilon\cos q_t\,\mathrm{d}W_t^1 + \varepsilon\sin q_t\,\mathrm{d}W_t^2 + \frac{\varepsilon^2}{2}\,\mathrm{d}t.$$

The stochastic integral $B_t = -\int_0^t \cos q_s\,\mathrm{d}W_s^1 + \int_0^t \sin q_s\,\mathrm{d}W_s^2$ has quadratic variation t and is a Brownian motion. Consequently

$$a_t = a_0\exp\left(\varepsilon B_t + \frac{\varepsilon^2}{2}t\right). \tag{6.2}$$

Since $B_t/t \to 0$ almost surely, (6.2) gives $t^{-1}\log(a_t/a_0) \to \varepsilon^2/2$. Thus $a_t \to \infty$ almost surely for every $\varepsilon > 0$. For any fixed $c > 0$,

$$\inf_y \|a_t e^{-|\cdot - q_t|} - c e^{-|\cdot - y|}\|_{H^1} = \sqrt{2}\,|a_t - c|.$$

Thus the solution retains its peakon shape while leaving every fixed-amplitude orbit, making $\mathcal{P}$ the natural comparison family under spatially varying momentum noise. Theorem 1.3 quantifies persistence near this family for perturbations with arbitrary nonnegative finite momentum, including diffuse measures, in terms of their initial H^1 velocity error.

6.2. Separation of two positive peakons

Proposition 6.1 (Separation under spatially constant noise)**.** *For spatially constant transport coefficients, there exist positive two-peakon initial data arbitrarily close in H^1 to a single peakon such that the peak separation tends to infinity and the distance to the single-peakon family has a strictly positive limit.*

Proof. With constant coefficients, the noise adds a common Brownian translation, while the deterministic equations for the weights and separation remain unchanged. Write $m = p_1\delta_{q_1} + p_2\delta_{q_2}$, $q_1 < q_2$, $r = q_2 - q_1 > 0$, and $p_1, p_2 > 0$. Then

$$\dot p_1 = -\tfrac{1}{2}p_1p_2e^{-r}, \quad \dot p_2 = \tfrac{1}{2}p_1p_2e^{-r}, \quad M = p_1 + p_2, \quad D = p_1p_2(1 - e^{-r})$$

with M, D constant and $0 < D < M^2/4$. The first weight decreases and the second increases, and both lie in $[0, M]$, so each has a limit. Since $p_1p_2 \geq D > 0$,

$$\int_0^\infty e^{-r(t)}\,\mathrm{d}t \leq \frac{2}{D}\big(p_1(0) - \lim_{t\to\infty} p_1(t)\big) < \infty.$$

Moreover, $|\dot r| \leq M/2$, so e^{-r} has a bounded derivative and is uniformly continuous. A nonnegative, integrable, uniformly continuous function tends to zero: otherwise disjoint intervals of a fixed positive length would contribute a fixed positive amount to its integral. Hence $r(t) \to \infty$. Passing to the limit in $D = p_1p_2(1 - e^{-r})$ shows that the limiting weights are the two distinct roots of $p(M - p) = D$. The limiting difference $p_2 - p_1$ cannot be negative, since $\dot r = (p_2 - p_1)(1 - e^{-r})/2$ would then be bounded above by a negative constant for all sufficiently large times. Since the roots are distinct by $D < M^2/4$, the limiting difference is positive. Hence

$$p_1 \longrightarrow p_- = \frac{M - \sqrt{M^2 - 4D}}{2} > 0, \quad p_2 \longrightarrow p_+ = \frac{M + \sqrt{M^2 - 4D}}{2}.$$

Between the two particles, $u_{xx} = u > 0$. Outside them, u tends monotonically to zero. Consequently

$$\max u = \frac{1}{2}\max\{p_1 + p_2e^{-r},\ p_2 + p_1e^{-r}\}, \quad E = \frac{1}{2}(p_1^2 + p_2^2 + 2p_1p_2e^{-r}).$$

The identity $\operatorname{dist}_{H^1}(u,\mathcal{P})^2 = E - 2(\max u)^2$ and the limits of p_1, p_2, r give

$$\lim_{t\to\infty} \operatorname{dist}_{H^1}(u_t,\mathcal{P})^2 = \frac{1}{2}p_-^2 > 0.$$

An arbitrarily small second initial weight gives initial data arbitrarily close in H^1 to a single peakon. □

The separating peak retains a positive limiting weight and hence a positive contribution to the global H^1 distance from $\mathcal{P}$. This describes the long-time effect of a separated component; the exit estimates in Theorem 1.3 quantify persistence near the family over their stated time intervals. For localized asymptotic peakon theory, see [34].

6.3. Particle computations and the comparison family

The computations focus on the comparison underlying Theorem 1.3: proximity to the full peakon family when the noise changes the height. Figure 1 combines the exact single-peakon amplitude law with a comparison of distances for perturbed data. Figure 2 then examines the initial defect estimate used in the proof and the resolution of the family distance and defect under time-step refinement. Throughout these experiments, the model is (1.1) with $\xi_1(x) = \sin x$, $\xi_2(x) = \cos x$, $\varepsilon = 0.3$ and $T = 2$. The same two Brownian motions drive every particle within a realization, and different Monte Carlo realizations are independent.

The particle scheme applies simultaneous Itô Euler updates to the first position, the logarithmic gaps and the logarithmic weights. Reconstructing positions from exponentiated gaps preserves particle order and positive weights in exact arithmetic. The time-step comparisons use common Brownian paths, with coarse increments obtained by summing fine increments. The accompanying supplementary numerical methods document provides the equations, random-number layout, seeds, interval formulas and diagnostic residuals. The `supplement/` directory contains the executable implementation, numerical arrays, machine-readable summaries and figure sources.

The reported confidence intervals are pointwise large-sample 95% Monte Carlo intervals that quantify sampling uncertainty without including discretization bias. RMS intervals are obtained by taking square roots of the intervals for the mean squared paired errors.

For the chosen sinusoidal fields, the exact single-peakon height law provides a first check of the amplitude update within the invariant family of Theorem 1.3(iii). With $a(0) = 1$ and $q(0) = 0$, (6.1)–(6.2) imply

$$\log a(T) \sim N(\varepsilon^2 T/2, \varepsilon^2 T) = N(0.09, 0.18).$$

Figure 1(a) compares the terminal histogram with this exact density. At $h = 1/512$, the simulated mean is 0.08906, with interval $[0.08003, 0.09808]$, and the sample variance is 0.17383, with the large-sample Gaussian variance interval $[0.16851, 0.17915]$. The sample mean amplitude is 1.19272, with interval $[1.18141, 1.20404]$, compared with the exact value $e^{0.18} = 1.19722$. The logarithmic Euler increment is an adapted orthogonal rotation of a Gaussian increment, so the Gaussian marginal law is retained exactly at each discrete time. The panel therefore provides a benchmark for the amplitude update; the refinement comparisons below examine the computed distances and defect.

To examine the comparison family in Theorem 1.3(i), take the three-particle perturbation

$$q(0) = (-1.5, 0, 1.5), \quad p(0) = (0.05, 2, 0.05).$$

With $e^{-|\cdot|}$ as the reference initial peakon, the initial H^1 error is 0.0512296, the initial family distance is 0.0487394, and $\mathcal{Z}(u_0) = 0.00213189$. For a positive particle measure, the maximum of u occurs at a particle position, since $u_{xx} = u > 0$ between consecutive particles. Thus the distance to $\mathcal{P}$ and the distance to the fixed-amplitude translation orbit are computed algebraically:

$$d_{\mathcal{P}}(u)^2 = E - 2(\max_i u(q_i))^2, \quad d_c(u)^2 = E + 2c^2 - 4c \max_i u(q_i), \quad c = 1.$$

Both distances are minimized over position, and $d_{\mathcal{P}}$ also allows the comparison height to vary. The difference between the two distances therefore records the effect of this additional optimization. The functional $\mathcal{Z}$ is evaluated from the eigenvalues of $T_{ij} = \sqrt{p_i p_j}\, e^{-|q_i - q_j|/2}$, using a factored difference to avoid cancellation near rank one; the supplementary notes give the evaluation formula.

At $T = 2$ and $h_{\mathrm{ref}} = 1/1024$, the mean family distance is 0.10025 with interval $[0.09777, 0.10272]$. The fixed-amplitude orbital distance has mean 0.62953 with interval $[0.60292, 0.65615]$. The mean of $\mathcal{Z}$ is 0.0040912, with a

95% interval of [0.0039233, 0.0042591]. The separation of the curves in Figure 1(b) makes this distinction visible over the displayed interval. Because both distances have already been minimized over translation, this comparison shows why the height is allowed to vary in the stability statement: Theorem 1.3(i) controls exit from a neighborhood of $\mathcal{P}$. The curves are ensemble medians, and the empirical central 90% bands describe variability across paths, rather than confidence intervals for a mean.

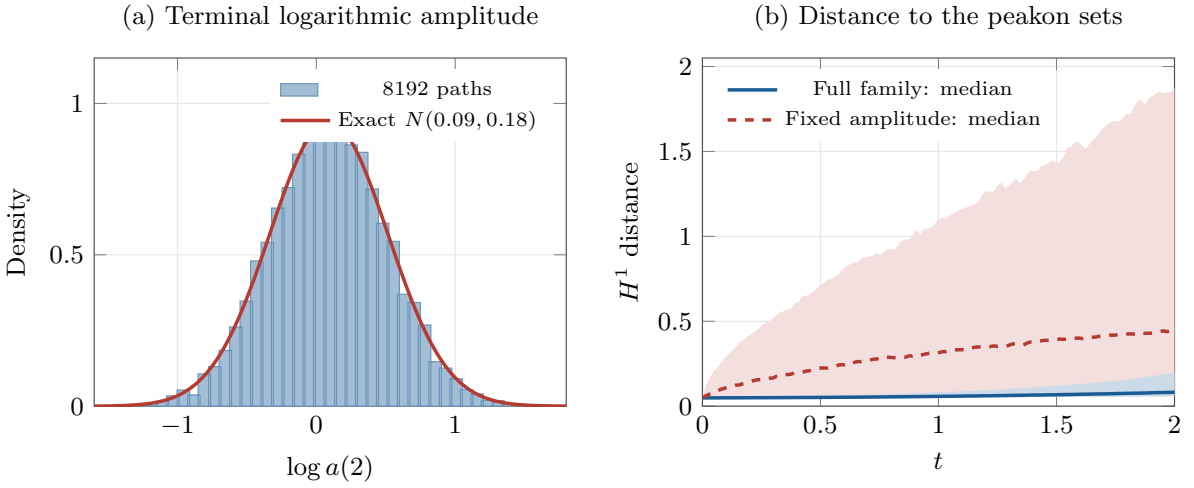


Figure 1: Amplitude variation and the comparison family in Theorem 1.3, for $\xi_1(x) = \sin x$, $\xi_2(x) = \cos x$, $\varepsilon = 0.3$ and $T = 2$. (a) Terminal logarithmic amplitudes from 8192 single-peakon realizations and the exact $N(0.09, 0.18)$ density, providing an amplitude benchmark within the invariant family. (b) Medians and pointwise 5–95% quantile bands from 2048 three-particle realizations, comparing $d_{\mathcal{P}}$ with the distance to the translation orbit of height 1. Both distances minimize over translation; the comparison height also varies in $d_{\mathcal{P}}$. The bands describe variability across realizations.

6.4. Initial defect scaling and refinement checks

A key step in the proof of Theorem 1.3(i) is the quadratic control of $\mathcal{Z}(u_0)$ in the initial H^1 error. Figure 2(a) illustrates this step for $p(0) = (r, 2, r)$ at the same three positions, using 25 logarithmically spaced values $0.002 \le r \le 0.2$. The initial H^1 error is exactly $\delta = r\sqrt{1 + e^{-3}}$. A log–log fit to the eight smallest errors gives a slope of 1.9999 for $\mathcal{Z}(u_0)$ as a function of δ. For this perturbation family, the initial-profile calculation thus displays the quadratic defect scaling used to obtain (1.5).

The refinement experiment checks the time-grid resolution of $d_{\mathcal{P}}$ and $\mathcal{Z}$, the distance and defect entering Theorem 1.3(ii). Table 1 and Figure 2(b) compare paired terminal values on common Brownian paths, using $h_{\mathrm{ref}} = 1/1024$ as the reference step. Both RMS differences decrease as the time step is refined, showing how the family distance and defect are resolved on finer time grids. The intervals quantify Monte Carlo uncertainty in the paired differences. These differences are measured against a finite numerical reference; establishing a strong convergence rate would require an error analysis against the exact solution. An additional Brownian-bridge refinement to $h = 1/2048$ checks the resolution of the reference computation and gives terminal RMS differences of 3.505×10^{-4} for $d_{\mathcal{P}}$ and 2.764×10^{-5} for $\mathcal{Z}$. Their respective 95% intervals are $[3.112, 3.858] \times 10^{-4}$ and $[2.548, 2.964] \times 10^{-5}$. The supplementary notes specify this coupling and the deterministic conservation checks.

h	RMS difference in $d_{\mathcal{P}}$, $\times 10^{-3}$	RMS difference in $\mathcal{Z}$, $\times 10^{-5}$
1/128	1.508 [1.317, 1.677]	11.057 [10.038, 11.990]
1/256	0.929 [0.811, 1.033]	6.847 [6.245, 7.401]
1/512	0.504 [0.424, 0.572]	3.925 [3.512, 4.298]

Table 1
Paired time-step refinement on 2048 common Brownian paths, at $T = 2$. Brackets give pointwise 95% Monte Carlo intervals; the reference step is 1/1024.

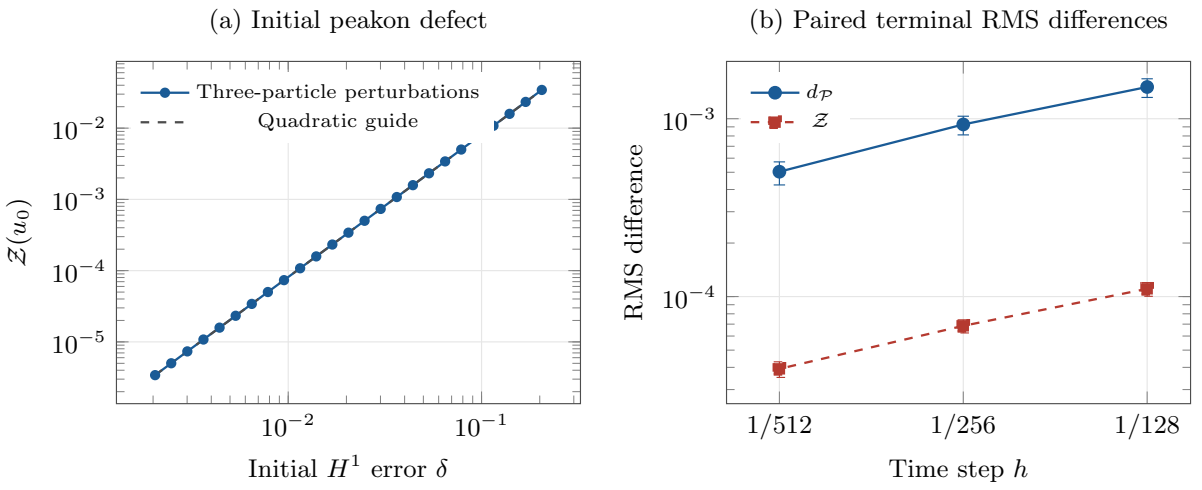


Figure 2: Initial defect control and resolution of the numerical observables. (a) The initial peakon defect $\mathcal{Z}(u_0)$ for positive three-particle perturbations $p(0) = (r, 2, r)$, with a quadratic guide anchored at the smallest perturbation. The scaling illustrates the initial defect control used in Theorem 1.3(i). (b) Paired terminal RMS differences in $d_{\mathcal{P}}$ and $\mathcal{Z}$ on 2048 common Brownian paths, relative to $h_{\mathrm{ref}} = 1/1024$. Their decrease under refinement checks the resolution of the computed distance and defect; the error bars are pointwise 95% Monte Carlo intervals.

Remark 6.2 (Interpretation of the numerical parameters). For these coefficients, $\kappa_{1,k} = 4$, $\kappa_{2,k} = 24$ and $C_Z = 1376$, so $C_Z\varepsilon^2 T = 247.68$. For the perturbed profile above and $\eta = 1$, the truncated bound in (1.6) is 1. In general, this estimate gives a probability smaller than one when

$$\mathcal{Z}(u_0) < \frac{\eta^4}{4\sqrt{2}} e^{-C_Z\varepsilon^2 T}.$$

The figures therefore illustrate the comparison geometry, the initial defect scaling and the resolution of the computed observables at the fixed parameters above. The quantitative exit estimates follow from the stochastic defect balance proved in Section 5; Corollary 1.6 then balances the quadratic initial error against its growth in time to obtain the logarithmically growing stability horizon.

A. Technical estimates for stochastic characteristics

This appendix supplies the three technical inputs to Section 4. The noise flow has global bounds for its displacement and first derivative, while its second derivative is controlled locally. The random-test identity removes the stochastic transport from the weak equation. The resulting characteristic representation then gives uniform weight bounds for the localized uniqueness estimate. Each step uses C_b^3 noise fields and requires no spatial moment of the initial momentum.

A.1. Spatial regularity of the noise flow

Let ϕ_t be the noise-only flow

$$\mathrm{d}\phi_t(x) = \varepsilon \sum_k \xi_k(\phi_t(x)) \circ \mathrm{d}W^k, \quad J_t(x) = \partial_x \phi_t(x), \quad a = \frac{\varepsilon^2}{2} \sum_k \xi_k \xi_k'.$$

The Itô drift of ϕ is a.

Lemma A.1. *The flow ϕ has a version that is twice continuously differentiable in space, with the derivatives locally jointly continuous in space and time. It consists of increasing diffeomorphisms and, on every finite interval,*

$$\sup_{t \le T} \left(\|\phi_t - \mathrm{id}\|_\infty + \|J_t\|_\infty + \|J_t^{-1}\|_\infty \right) < \infty \quad \textit{almost surely}. \tag{A.1}$$

Here J_t^{-1} means the reciprocal Jacobian. The second spatial derivative of the flow is bounded on compact space–time sets; a global bound on that derivative is not asserted.

Proof. The first variation satisfies

$$\mathrm{d}J = a'(\phi) J \, \mathrm{d}t + \varepsilon \sum_k \xi_k'(\phi) J \, \mathrm{d}W^k, \quad J_0 = 1.$$

A difference-quotient argument and Itô–BDG–Gronwall estimates give, for every finite $p \ge 2$,

$$\begin{aligned} \sup_x \mathbb{E} \sup_{t \le T} |J_t(x)|^p < \infty, \\ \mathbb{E} \sup_{t \le T} |\phi_t(x) - \phi_t(y)|^p + \mathbb{E} \sup_{t \le T} |J_t(x) - J_t(y)|^p \le C_{p,T} |x - y|^p. \end{aligned} \tag{A.2}$$

The estimate for J uses the boundedness of a'' and ξ_k'', which makes the coefficients a' and ξ_k' Lipschitz.

The second variation H solves

$$\mathrm{d}H = \left(a''(\phi) J^2 + a'(\phi) H \right) \mathrm{d}t + \varepsilon \sum_k \left(\xi_k''(\phi) J^2 + \xi_k'(\phi) H \right) \mathrm{d}W^k, \quad H_0 = 0.$$

This equation identifies the derivative of J in $L^p(\Omega; C([0,T]))$. To justify the identification, take difference quotients in the equation for J and apply the mean-value formula to a' and ξ_k'. Since a'' and ξ_k'' are bounded and continuous, localizing ϕ to compact sets allows us to use their uniform continuity there, while higher moments justify dominated convergence. The linear SDE estimate then gives convergence of the difference quotients to H, locally in the initial-label variable, without a Lipschitz assumption on a''.

A continuous spatial version follows from the identity

$$\frac{H_t(x)}{J_t(x)} = \int_0^t \left(a'' - \varepsilon^2 \sum_k \xi_k' \xi_k'' \right) (\phi_s(x)) J_s(x) \, \mathrm{d}s + \varepsilon \sum_k \int_0^t \xi_k''(\phi_s(x)) J_s(x) \, \mathrm{d}W_s^k.$$

The first integral is pathwise jointly continuous on compact space–time sets. For the stochastic integrand, the spatial L^p increments are of order $|x - y|$, by (A.2) and boundedness of ξ_k'''. The BDG inequality gives the same order in space for the stochastic integral, together with temporal increments of order $|t - s|^{1/2}$ in L^p. Kolmogorov's criterion with arbitrarily large p therefore yields a jointly continuous version of H. The linear equation for H, boundedness of a'' and ξ_k'', and the higher moments of J also give

$$\sup_x \mathbb{E} \sup_{t \le T} |H_t(x)|^p < \infty \quad (p \ge 2).$$

The estimates at a higher exponent give uniform integrability. Together with joint pathwise continuity, this shows that $x \mapsto H(x)$ is continuous as a map into $L^p(\Omega; C([0,T]))$. Together with the difference-quotient identification above, this makes $x \mapsto J(x)$ continuously differentiable in this Banach space. The fundamental theorem of calculus in

$L^p(\Omega; C([0,T]))$ gives $J_t(y) - J_t(x) = \int_x^y H_t(z)\,\mathrm{d}z$. First take rational endpoints and then use continuity to obtain this identity pathwise for all labels and times. Consequently $H = \partial_x^2\phi$ is the classical second spatial derivative.

It remains to prove spatially uniform bounds; these do not follow by taking a supremum in (A.2). Fix $1/3 < \alpha < \beta < 1/2$ and a β-Hölder version of the Brownian Stratonovich rough path $\mathbf{W}$. The rough differential equation with vector fields $\epsilon\xi_k$ has a global solution for every initial label under the C_b^3 assumption. For each fixed label, the solution is adapted because its restriction to $[0,t]$ depends only on the enhanced Brownian path restricted to $[0,t]$. The compensated sums defining its rough integral have a first-level part converging to the Itô integral. Their Itô iterated-integral part tends to zero in L^2, and their diagonal part converges to

$$\frac{\epsilon^2}{2}\sum_k \int_0^t (\xi_k\xi_k')(\phi_s(x))\,\mathrm{d}s.$$

Indeed, the coefficients in the second-level sums are bounded and adapted, and the variance of the sum of their Itô-area terms is bounded by a constant times the sum of the squared lengths of the partition intervals. The rough solution therefore solves the noise-only Itô SDE. Pathwise uniqueness identifies it with the stochastic flow at each fixed label. A countable dense set of labels and spatial continuity yield this identification simultaneously for all labels and times.

The RDE estimates in [20, Proposition 8.2 and Theorem 8.3] give, uniformly over the initial label,

$$\begin{aligned}&\sup_{x,t\le T}|\phi_t(x)-x| \le C, \quad |\phi_t(x)-\phi_s(x)| \le C|t-s|^\alpha,\\ &\phi_t(x)-\phi_s(x) = \epsilon\sum_j \xi_j(\phi_s(x))W_{s,t}^j + R_{s,t}^\phi(x), \quad \sup_x|R_{s,t}^\phi(x)| \le C|t-s|^{2\alpha}.\end{aligned} \tag{A.3}$$

The local controlled-path estimates depend on bounds for the vector fields and their derivatives and are independent of the initial point. Combining the estimates on successive control intervals preserves this uniformity.

The integrand $\xi_k'(\phi(x))$ is controlled by W with derivative $\epsilon\xi_k''(\phi(x))\xi_j(\phi(x))$. Its controlled norm is uniform in x, since boundedness of ξ_k''' controls the composition remainder and gives the Hölder bound for the controlled derivative. Controlled composition and integration [20, Lemma 7.3 and Theorem 4.10] therefore give

$$\sup_{x,t\le T}\left|\epsilon\sum_k\int_0^t \xi_k'(\phi_s(x))\,\mathrm{d}\mathbf{W}_s^k\right| \le C.$$

For each label, the rough integral agrees with the Stratonovich stochastic integral by the adapted compensated-sum argument given below for the represented characteristics. Itô's formula gives

$$\log J_t(x) = \epsilon\sum_k\int_0^t \xi_k'(\phi_s(x))\circ\,\mathrm{d}W_s^k.$$

Passing from rational labels to all labels by continuity shows that J and its reciprocal are bounded uniformly in space and time. The estimates also give uniform temporal continuity of J. Together, bounded displacement and positivity of J show that ϕ_t is an increasing diffeomorphism onto $\mathbb{R}$. All constants in (A.3) and (A.1) are deterministic on bounded rough-path norm balls. □

A.2. A random test function and commutator estimates

For $f \in C_c^\infty$, set $\psi_t = (J_t f)\circ\phi_t^{-1}$, which is a compactly supported C^1 random field whose local C^1 norm varies continuously. For a deterministic $g \in C_c^\infty$, a change of variables gives

$$\int_{\mathbb{R}}\psi_t(z)g(z)\,\mathrm{d}z = \int_{\mathbb{R}} f(x)J_t(x)^2 g(\phi_t(x))\,\mathrm{d}x.$$

Applying Itô's formula to $J^2 g(\phi)$, followed by stochastic Fubini on the fixed compact support of f, gives

$$\mathrm{d}\psi_t = \frac{\epsilon^2}{2}\sum_k B_k^2\psi_t\,\mathrm{d}t - \epsilon\sum_k B_k\psi_t\,\mathrm{d}W^k \tag{A.4}$$

as an identity of distributions: the operator on g is $\xi_k g' + 2\xi_k' g = -B_k^* g$, and its square gives the Itô correction. This formulation requires neither a pointwise semimartingale equation for ψ nor a function-valued ψ_{xx}.

Let P_δ be convolution with a smooth compactly supported mollifier ρ_δ.

Lemma A.2. *For $B = \xi\partial_x - \xi'$, $\xi \in C_b^3$, and $\psi \in C_c^1$,*

$$\|[B, P_\delta]\psi\|_\infty + \|[B, [B, P_\delta]]\psi\|_\infty \le C_{\xi,\rho}\delta\|\psi\|_{C^1}, \quad 0 < \delta \le 1. \tag{A.5}$$

Proof. Put $r = x - y$, $h = \xi(x) - \xi(y)$, $d = 2\xi'(y) - \xi'(x)$, and $w = \xi(x)\xi'(x) - \xi(y)\xi'(y)$. Integration by parts gives the first commutator kernel $h\rho_\delta'(r) + d\,\rho_\delta(r)$. Commuting once more gives

$$h^2\rho_\delta''(r) + (w + 2dh)\rho_\delta'(r) + \big(2\xi(y)\xi''(y) - \xi(x)\xi''(x) + d^2\big)\rho_\delta(r).$$

Both kernels have uniformly bounded $L^1(\,\mathrm{d}y)$ norms. For the second kernel, write $b = \xi'(x)$ and $c = \xi(x)\xi''(x)$. Its leading terms at $y = x$ are

$$b^2r^2\rho_\delta''(r), \quad (3b^2 + c)r\rho_\delta'(r), \quad (b^2 + c)\rho_\delta(r).$$

These terms cancel after integration because their moments are $2, -1, 1$. The error coefficients are respectively $O(|r|^3)$, $O(r^2)$ and $O(|r|)$, uniformly in x under the C_b^3 assumption, so their contribution after integration is $O(\delta)$. By the uniform kernel bound, replacing $\psi(y)$ by $\psi(x)$ introduces an error bounded by $C\delta\|\psi'\|_\infty$. This proves the second bound. The leading kernel of the first commutator is $b(r\rho_\delta' + \rho_\delta)$, whose integral also vanishes, so the same argument applies. □

Fix $\delta > 0$ and stop both m and ψ when the mass, the support radius of ψ, or its C^1 norm reaches a deterministic bound. On the stopped interval, set

$$z = P_\delta\psi, \quad a_\delta = \frac{\varepsilon^2}{2}\sum_k P_\delta B_k^2\psi, \quad c_{\delta,k} = -\varepsilon P_\delta B_k\psi.$$

The distributional identity (A.4) gives $\mathrm{d}z = a_\delta\,\mathrm{d}t + \sum_k c_{\delta,k}\,\mathrm{d}W^k$. All these fields are smooth, supported in one deterministic compact interval, and have deterministic bounds on each spatial derivative at this fixed δ. For the coefficient involving $B_k^2\psi$, derivatives may be transferred to the convolution kernel in the distributional pairing.

Take deterministic finite-rank operators S_n obtained by Fourier projection on a larger interval followed by multiplication by a smooth cutoff equal to one on the common support. The interval is chosen so that all fields vanish near its endpoints. Their spatial derivative bounds imply, uniformly on the stopped interval,

$$S_nz \to z, \quad S_na_\delta \to a_\delta, \quad S_nc_{\delta,k} \to c_{\delta,k}$$

in C^2. The Fourier coefficients of S_nz are scalar semimartingales, with coefficients obtained by applying the same deterministic operators to a_δ and $c_{\delta,k}$. We apply the ordinary product rule to these coefficients, use the deterministic weak tests defining S_n, and pass to the limit as $n \to \infty$ to obtain

$$\mathrm{d}\langle z, m\rangle = \left[F_z(m) + \frac{\varepsilon^2}{2}\sum_k\langle B_k^2z, m\rangle + \langle a_\delta, m\rangle + \varepsilon\sum_k\langle B_kc_{\delta,k}, m\rangle\right]\mathrm{d}t + \sum_k\left[\varepsilon\langle B_kz, m\rangle + \langle c_{\delta,k}, m\rangle\right]\mathrm{d}W^k.$$

These limits follow from the stopped mass bound and Itô isometry, with the drift controlled in the C^1 norm and the correction in the C^2 norm. With all integrals stopped at the same time, substituting $z, a_\delta, c_{\delta,k}$ yields

$$\mathrm{d}\langle P_\delta\psi, m\rangle = F_{P_\delta\psi}(m)\,\mathrm{d}t + \varepsilon\sum_k\langle [B_k, P_\delta]\psi, m\rangle\,\mathrm{d}W^k + \frac{\varepsilon^2}{2}\sum_k\langle [B_k, [B_k, P_\delta]]\psi, m\rangle\,\mathrm{d}t. \tag{A.6}$$

Before passing to the smoothing limit, combine the second-order terms:

$$B_k^2P_\delta + P_\delta B_k^2 - 2B_kP_\delta B_k = [B_k, [B_k, P_\delta]].$$

The localization is justified because the mass is continuous, and the support radius and C^1 norm of ψ are adapted and continuous. Indeed, on the fixed support of f,

$$\psi_t(\phi_t(x)) = J_t(x)f(x), \quad \psi_t'(\phi_t(x)) = f'(x) + \frac{H_t(x)}{J_t(x)}f(x).$$

The local second-derivative bounds control these quantities. For fixed δ, the higher derivative bounds of the smoothed fields may depend on δ; they are only used before the smoothing limit. Estimate (A.5) and the stopped BDG inequality show that the stochastic error in (A.6) tends to zero; the drift error vanishes as well. Since $P_\delta\psi \to \psi$ in C^1 on the localized compact sets and $|F_\psi(m)| \le C\|\psi\|_{C^1} M^2$, the nonlinear drift converges. Removing the stops gives

$$\mathrm{d}\langle \psi, m\rangle = F_\psi(m)\,\mathrm{d}t. \tag{A.7}$$

Applying the identity first to a countable dense family of f and then using the localized operator bounds yields a single event on which it holds for every compactly supported smooth f.

A.3. Transport with a locally bounded reaction

Lemma A.3 (Locally bounded reaction)**.** *Let v be continuous and bounded on $[0,T]\times\mathbb{R}$ for every finite T, and locally Lipschitz in space uniformly on compact time intervals. Let b be a locally bounded Borel function. Every finite narrowly continuous nonnegative measure solution of $\partial_t \nu + \partial_x(v\nu) = b\nu$ has the representation*

$$\nu_t = (Q_{0,t})_{\#}(r_t \nu_0), \quad r_t(x) = \exp\left(\int_0^t b_s(Q_{0,s}(x))\,\mathrm{d}s\right), \tag{A.8}$$

where Q is the characteristic flow of v.

Proof. The bounded speed gives a global characteristic flow with bounded displacement. For a fixed finite time interval and a compact terminal support, all corresponding backward characteristics remain in a fixed larger compact set. Extend v outside this set to a bounded globally Lipschitz field and truncate b outside a still larger set. Choose the extensions with a uniform speed bound, so that the backward supports remain where the extended and original coefficients agree.

Mollify this extended v spatially and use the smooth backward tests $f^n_s = f\circ Q^n_{s,t}$. Their supports and first derivatives have uniform bounds; the transport residual is $(v - v^n)\partial_x f^n_s$, which tends uniformly to zero. Time-dependent tests are justified by approximation in the weak identity. Passing to the limit gives

$$\langle f, \nu_t\rangle = \langle f\circ Q_{0,t}, \nu_0\rangle + \int_0^t \langle b_s(f\circ Q_{s,t}), \nu_s\rangle\,\mathrm{d}s.$$

The Borel coefficient b is retained without spatial approximation. The identity extends to continuous compactly supported terminal tests by uniform approximation. Put $\eta_t = (Q^{-1}_{0,t})_{\#}\nu_t$ and $c_t(x) = b_t(Q_{0,t}(x))$. For $g \in C_c(\mathbb{R})$, choose the terminal test $f = g\circ Q^{-1}_{0,t}$. The flow identity $Q^{-1}_{0,t}\circ Q_{s,t} = Q^{-1}_{0,s}$ gives

$$\langle g, \eta_t\rangle = \langle g, \nu_0\rangle + \int_0^t \langle c_s g, \eta_s\rangle\,\mathrm{d}s.$$

On every compact interval I of initial labels, c is bounded on $[0,T]\times I$. The right-hand side defines a finite signed measure on each compact interval of labels: for a Borel set $A \subset I$, its integral term is $\int_0^t \int_A c_s(x)\eta_s(\mathrm{d}x)\,\mathrm{d}s$. Equality on continuous compactly supported tests therefore implies equality on all such Borel sets. In particular,

$$\eta_t|_I = \nu_0|_I + \int_0^t c_s\eta_s|_I\,\mathrm{d}s.$$

Fubini's theorem shows that $r_t\nu_0|_I$, with the stated exponential weight, satisfies the same identity. If μ_t is their difference and $C_I = \sup_{[0,T]\times I}|c|$, then

$$\|\mu_t\|_{\mathrm{TV}(I)} \le C_I \int_0^t \|\mu_s\|_{\mathrm{TV}(I)}\,\mathrm{d}s.$$

The total-variation norms are measurable and locally bounded; one may use a countable dense subset of the unit ball of $C(I)$ to express the norm. Gronwall's inequality gives $\mu_t = 0$. Exhausting $\mathbb{R}$ by compact intervals proves (A.8). The represented measure is finite since it equals ν_t, so no a priori global integrability of the weight is needed. The actual Borel values of b are retained throughout, so the argument includes atoms. □

A.4. Uniform bounds for the characteristic weights

Proof of Lemma 4.1. Fix T and use the representation from Section 4. After localizing the mass and the α rough-path norm of W, the estimates in Lemma A.1 have deterministic constants. Since $|\dot{Q}| \le M/(2J(Q))$, Q is Lipschitz in time with a bound independent of its initial label. Decompose

$$q_t(x) - q_s(x) = \phi_t(Q_t(x)) - \phi_t(Q_s(x)) + \phi_t(Q_s(x)) - \phi_s(Q_s(x)).$$

The first term is $O(|t-s|)$ uniformly in the label, using the global bound for J_t. By (A.3), the second is $\varepsilon \sum_j \xi_j(q_s(x)) W^j_{s,t} + O(|t-s|^{2\alpha})$. Since $2\alpha < 1$, the first term is also a controlled remainder. Thus the controlled increment norms of $q(x)$ are uniform in the label; this does not assert a uniform bound on the initial value $q_0(x) = x$.

The function $\xi'_k(q(x))$ is controlled with derivative $\varepsilon \xi''_k(q(x))\xi_j(q(x))$. Its controlled norm is uniform by boundedness of ξ'''_k. The rough integral estimate therefore gives

$$\sup_{x,t\le T} \left| \varepsilon \sum_k \int_0^t \xi'_k(q_s(x))\, \mathrm{d}\mathbf{W}^k_s \right| \le C.$$

For each fixed label, this rough integral agrees with the Stratonovich integral without further flow regularity. Write $Y_k = \xi'_k(q)$ and $Y'_{k,j} = \varepsilon \xi''_k(q)\xi_j(q)$. Both are adapted, and Y' is bounded. The compensated sums defining the rough integral are

$$\sum_{[s,t]\in\pi} \sum_k Y_k(s) W^k_{s,t} + \sum_{[s,t]\in\pi} \sum_{j,k} Y'_{k,j}(s) \mathbb{W}^{j,k}_{s,t}.$$

For the Brownian Stratonovich lift, $\mathbb{W}^{j,k}_{s,t}$ is the Itô iterated integral plus $\frac{1}{2}\delta_{jk}(t-s)$. The sum of the Itô-area terms converges to zero in L^2: its second moment is bounded by $C\sum_{[s,t]\in\pi}(t-s)^2$, using boundedness and adaptedness of Y'. The first sum converges to the Itô integral, and the diagonal terms converge to $\frac{1}{2}\int_0^{\cdot} \sum_k Y'_{k,k}(s)\,\mathrm{d}s$. Since the characteristic Itô equation gives $[Y_k, W^k]_t = \int_0^t Y'_{k,k}(s)\,\mathrm{d}s$, this is exactly the Stratonovich integral. The continuous versions agree for all times almost surely. Consequently, for m_0-almost every label on a single probability-one event,

$$\log p_t(x) = -\int_0^t \bar{u}_{s,x}(q_s(x))\,\mathrm{d}s - \varepsilon \sum_k \int_0^t \xi'_k(q_s(x))\,\mathrm{d}\mathbf{W}^k_s.$$

The first term is bounded by $\frac{1}{2}\int_0^T M_s\,\mathrm{d}s$. This proves the claimed essential bounds on p and p^{-1}. Fubini's theorem ensures that these bounds hold on a common event of probability one, without requiring continuity of p in the label or an extension from rational labels for p.

To define the localizing stopping times, use the running α rough-path seminorm with $\alpha < \beta < 1/2$. It is adapted and continuous: the β-Hölder bound makes the near-diagonal α seminorm vanish uniformly when the interval shrinks. Deterministic estimates on rough-path norm balls, together with the mass bounds, then control the weights without taking a supremum over an arbitrary Borel weight field.

□

Declarations

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Data availability. The code, fixed random seeds, simulation outputs and figure sources for Section 6, together with supplementary numerical methods and reproduction instructions, are provided in the accompanying `supplement/` directory.

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